Kinetic Theory

Lecture by Dr. Nikolai Leopold

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Disclaimer

These notes were taken during the lecture *Kinetic Theory* at the university of Basel given by Dr. Nikolai Leopold. Due to the complexity and the intricacy of the subject and the fact that there is not a lot of time in one lecture, these notes will be full of typos and errors and are thus to be read carefully.

Contents

1. Introduction

1.1. Motivation

Our goal will be to model the behaviour of gas (or systems made up of many particles) by a distribution function in the particle phase space. As gas is made up of an extremely large quantity of particles we will, for modelling purposes, think of it as a continuum.

Different simplified gas descriptions can and will lead to different equations for the above distribution functions $f : [0,T] \times \mathbb{R}^d \to \mathbb{R}^d$. In *classical mechanics* we would look at the motion of each particle separately which is too complex for our case. Thus we seek to find a simpler description describing the collective behaviour of the system.

Density on $\mathbb{R}^3 \times \mathbb{R}^3$ describes only important physical quantities on a microscopic scale (think of it as looking from far away).

An interpretation of f: One could think of f as an approximation for the true density of the gas in the phase space on a scale which is much larger than the typical distances between particles. For example: Consider some particles in one dimension. Then put imaginary boxes of width $\delta > 0$ around those particles. As $\delta \to 0$ the approximation of the behaviour will get better.

Note that the numbers of particles in $\Lambda \subseteq \mathbb{R}^d \times \mathbb{R}^d$ at time t is given by

$$\int_\Lambda f(x,t,v)\,\mathrm{d}x\,\mathrm{d}v \qquad \text{where } f(t,\cdot,\cdot) \in L^1_{\text{loc}}(\mathbb{R}^d \times \mathbb{R}^d).$$

The free transport equation reads as

$$(\partial_t + v \cdot \nabla_x) f(t,x,v) = 0$$

where free means that there is no external force. If there was external force we would have

$$(\partial_t + v \cdot \nabla_x + \underbrace{F(t,x)}_{\text{external Force}} \cdot \nabla_x) f(t,x,v) = 0$$

which is a more general transport equation. For example we can take a conservative force $F = -\nabla_x \Phi$ with $\Phi : \mathbb{R}^d \to \mathbb{R}^d$. In this case there is no interaction between the particles.

If however the force F is dependent on f (which models interaction between the particles) then we obtain the so-called *Vlasov equation*:

$$\begin{cases} (\partial_t + v \cdot \nabla_x + F(t,x) \cdot \nabla_v) f = 0 \\ F(t,x) = (-\nabla_x \Phi \star \rho(t,\cdot))(x) \end{cases} \qquad \text{where } \rho(t,x) = \int_{\mathbb{R}^d} f(t,x,v)\,\mathrm{d}v\,.$$

A further case of a modified transport equation is the so called *Bolzmann equation* which models collisions in particles:

$$(\partial_t + v \cdot \nabla_x) f = Q(f,f)$$

where the term $Q(f,f)$ is a collision kernel which is given by an integral term. We thus speak here of an integro differential equation.

In this lecture we will deal with the following topics:

- Transport equation
- Vlasov equation
- Vlasov-Poisson equation
- derivation of Vlasov equation from a model of N interacting particles.

1.2. A first example

Let $v \in \mathbb{R} \setminus \{0\}$ and consider the equation

$$\begin{cases} 0 = (\partial_t + v\partial_x) f(t,x) = \Bigg(\underbrace{\begin{bmatrix} 1 \\ v \end{bmatrix}}_{:=\vec{V}} \cdot \begin{bmatrix} \partial_t \\ \partial_x \end{bmatrix} \Bigg) f(t,x) \\ \\ f(0,\cdot) = f^{\text{in}} \in C^1(\mathbb{R}) \end{cases} \qquad x,t \in \mathbb{R} \tag{1.1}$$

where the operator $(\partial_t + v\partial_x)$ is the directional derivative in the direction of $\vec{V}$. Moreover consider the system of characteristics:

$$\begin{cases} \dot{\gamma}(t) = v \\ \gamma(0) = x_0 \end{cases} . \tag{1.2}$$

We find the unique solution to this characteristics system by integrating:

$$\gamma(t) = \gamma(0) + \int_0^t \dot{\gamma}(s)\,\mathrm{d}s = x_0 + tv.$$

We now propose the following:

Proposition 1.1. There exists a *unique* solution to (1.1).

Proof. We start with the uniqueness: Assume there exist a solution f to (1.1). Then

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} f(t,\gamma(t)) &= (\partial_t f)(t,\gamma(t)) + (\partial_x f)(t,\gamma(t)) \underbrace{\dot{\gamma}(t)}_{=v} \\ &= (\partial_t + v \cdot \partial_x) f(t,\gamma(t)) \overset{(1.1)}{=} 0. \end{aligned}$$

Moreover, if we define $x := x_0 + tv$ we have $f(t, x_0 + tv) = f(0, x_0)$ which implies that $f(t,x) = f(0, x - tv)$. Since any solution to (1.1) is of that form we obtain uniqueness of the solution to (1.1).

In order to prove the existence take an initial function $f^{\text{in}} \in C^1(\mathbb{R})$ and set $f(t,x) = f^{\text{in}}(x - tv)$. Then f solves (1.1). Thus we deduce that the Cauchy problem

$$\begin{cases} (\partial_t + v\partial_x) f(t,x) = 0 \\ f(0,\cdot) = f^{\text{in}} \in C^1(\mathbb{R}) \end{cases}$$

has a unique solution thus proving the claim. $\square$

Note that (1.1) is called the *Eulerian picture* and (1.2) is called *Lagrangian picture.*

2. The Transport equation: The classical setting

2.1. The Flow of a Vector Field

Consider the equation

$$0 = (\partial_t + v \cdot \nabla_x + F(t,x) \cdot \nabla_v) f(t,x,v) = \left(\partial_t + \underbrace{\begin{bmatrix} v \\ F(t,x) \end{bmatrix}}_{=:b(t,x,v)\in\mathbb{R}^{2d}} \cdot \begin{bmatrix} \nabla_x \\ \nabla_v \end{bmatrix}\right) f(t,x,v).$$

Grouping x and v into a single variable $z = (x,v)$ suggests to study the more general equation

$$\partial_t u(t,z) + b(t,z) \cdot \nabla_z u(t,z) = 0 \in \mathbb{R}^{2d}.$$

In the following we will use the methods of characteristics for the transport equation. The method consists of two steps:

(1) Define the flow of characteristics

(2) Use the flow to solve the transport equation.

We start with the characteristics flow:

Theorem 2.1. Let $T > 0$ and $b \in C^1([0,T] \times \mathbb{R}^d; \mathbb{R}^d)$ be bounded. Then for any $(t,x) \in [0,T] \times \mathbb{R}^d$ the ODE

$$\begin{cases} \dot\gamma(s) = b(s,\gamma(s)) \\ \gamma(t) = x \end{cases}$$

has a unique C^1 solution γ.

Proof. Since b is C^1 we have by the fundamental theorem of calculus that

$$\begin{aligned} b(s,y_1) - b(s,y_2) &= \int_{\lambda=0}^1 \frac{\mathrm{d}}{\mathrm{d}\lambda} b(s,(y_2-y_1)\lambda + y_1)\,\mathrm{d}\lambda \\ &= \int_{\lambda=0}^1 \left(\sum_{j=1}^d (\partial_{x_j} b)(s,(y_2-y_1)\lambda + y_1)(y_2-y_1)^j\right)\mathrm{d}\lambda\,. \end{aligned}$$

Then for any $(s,y_1),(s,y_2) \in V := [0,T] \times \overline{B}_\delta(x_0)$ for any $\delta > 0$ we obtain

$$|b(s,y_1) - b(s,y_2)| \le d \sup_{\substack{(s,y)\in V \\ j=1,\dots,d}} |\partial_{x_i} b(s,y)| \cdot |y_1 - y_2| < +\infty$$

because we are taking the supremum of a continuous function over a compact intervall.

We thus have proved that b is locally uniformly Lipschitz in the second variable y. By Picard-Lindelöf this implies that there exists a unique bounded C^1 solution. This solution can be extended to $[0,T] \times \mathbb{R}^d$ by the fact that b is a bounded vector field. $\square$

This motivates the following definition:

Definition 2.2. Let $T > 0$ and $b \in C^1([0,T] \times \mathbb{R}^d; \mathbb{R}^d)$ be bounded. Thanks to theorem 2.1 we find a unique function $X : [0,T] \times [0,T] \times \mathbb{R}^d \to \mathbb{R}$ satisfying

$$\begin{cases} \dfrac{\partial}{\partial s} X(s,t,x) = b(s, X(s,t,x)) \\ X(t,t,x) = x \end{cases} \quad \text{for all } t, s \in [0,T], x \in \mathbb{R}^d.$$

We call this function X the **flow of** b. It can be easily checked that X satisfies

$$\begin{cases} X \in C^1([0,T] \times [0,T] \times \mathbb{R}^d; \mathbb{R}^d) \\ \partial_s \partial_{x_j} X(s,t,x) = \partial_{x_i} \partial_s X(s,t,x) \text{ exist for any } s,t,x \text{ and are } C^0 \end{cases}.$$

Theorem 2.3. Let $T > 0$, $b \in C^1([0,T] \times \mathbb{R}^d; \mathbb{R}^d)$ bounded and let X be the flow of b as seen in definition 2.2.

(1) Then X satisfies
$$X(t_3, t_2, X(t_2, t_1, x)) = X(t_3, t_1, x)$$
for any $x \in \mathbb{R}^d, t_i \in [0,T], i = 1,2,3$. We call this the *flow property.*

(2) For any $s,t \in [0,T]$, $X(s,t,\cdot) : x \in \mathbb{R}^d \mapsto X(s,t,x) \in \mathbb{R}^d$ is a C^1 diffeomorphism i.e., the inverse of this function exists and is also continuously differentiable.

(3) We define the Jacobian of X by
$$J(s,t,x) := \det\left(\nabla_x X(s,t,x)\right),$$
where $\nabla_x X$ denotes the Jacobian matrix of $x \mapsto X(s,t,x)$ for fixed s,t. Then $J(s,t,x) > 0$ for any $(s,t,x) \in [0,T] \times [0,T] \times \mathbb{R}^d$ and satisfies the Cauchy problem
$$\begin{cases} \partial_s J(s,t,x) = \operatorname{div}_x b(s, X(s,t,x)) J(s,t,x) & \text{for any } s,t \in [0,T], x \in \mathbb{R}^d \\ J(t,t,x) \equiv 1 \end{cases}.$$

Proof. (1) Let $t_1, t_2 \in [0,T], x \in \mathbb{R}^d$ and define $\gamma, \tilde{\gamma} : [0,T] \to \mathbb{R}^d$ by

$$\gamma(t_3) = X(t_3, t_1, x), \quad \tilde{\gamma}(t_3) = X(t_3, t_2, X(t_2, t_1, x))$$

Both of those functions solve
$$\begin{cases} \dot{\gamma}(t) = b(t, \gamma(t)) \\ \gamma(t_2) = X(t_2, t_1, x) \end{cases}.$$
Since the solution of this ODE is unique it follows that $\tilde{\gamma} = \gamma$ i.e., X has the flow property as stated in the theorem.

(2) By the first part, $X(s,t,\cdot) \circ X(t,s,\cdot) = X(s,s,\cdot) \overset{\text{Def. 2.2}}{=} \text{id}$ which proves that the map $X(s,t,\cdot)$ is bijective. By Theorem 2.1, $X(s,t,\cdot) \in C^1$ we thus have proved the second point.

(3) By differentiating the equation

$$X(s,t,X(t,s,x)) = x$$

we obtain

$$I = ((\nabla_x X)(s,t,X(t,s,x)) \cdot (\nabla_x X)(t,s,x)$$

where I denotes the identity matrix. Taking determinats yields

$$1 = J(t,s,x) \cdot J(s,t,X(t,s,x)).$$

This implies that $J(t,s,\cdot)$ is never zero and thanks to the continuity it cannot change signs. Together with $J(t,t,x) = \det(I) = 1$ we see that $J(s,t,\cdot)$ is a strictly positive function, which proves the first part of the last point. To prove that J solves the Cauchy problem recall Jacobi's formula: for $A \in C^1([0,T], \mathrm{GL}_d(\mathbb{R}))$ we have that

$$\frac{\mathrm{d}}{\mathrm{d}s} \det(A(s)) = \det(A(s)) \operatorname{tr}\left\{A(s)^{-1} \cdot \frac{\mathrm{d}}{\mathrm{d}s} A(s)\right\},$$

where $A(s)^{-1}$ denotes the inverse matrix of $A(s)$. Using it on our problem we obtain (using the Schwartz theorem to swap derivatives in the first line ∂_s and div_x),

$$\begin{aligned}
\frac{\partial}{\partial s} J(s,t,x) &= \frac{\partial}{\partial s} \det\Big((\nabla_x X)(s,t,x)\Big) \\
&= J(s,t,x) \cdot \operatorname{tr}\left\{\big[(\nabla_x X)(s,t,x)\big]^{-1} \cdot \frac{\partial}{\partial s}(\nabla_x X)(s,t,x)\right\} \\
&\overset{\frac{\partial}{\partial s}X=b}{=} J(s,t,x) \cdot \operatorname{tr}\left\{\big[(\nabla_x X)(s,t,x)\big]^{-1} \cdot \nabla_x\big(b(s,X(s,t,x))\big)\right\} \\
&= J(s,t,x) \cdot \operatorname{tr}\left\{\big[(\nabla_x X)(s,t,x)\big]^{-1} \cdot (\nabla_x b)(s,X(s,t,x)) \cdot (\nabla_x X)(s,t,x)\right\} \\
&\overset{(!)}{=} J(s,t,x) \cdot \operatorname{tr}\left\{(\nabla_x b)(s,X(s,t,x))\right\},
\end{aligned}$$

where at (!) we used that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Since

$$\operatorname{tr}\left\{(\nabla_x b)(s,X(s,t,x))\right\} = (\operatorname{div}_x b)(s,X(s,t,x))$$

the claim follows. This concludes the proof.

□

2.2. Solving the Transport Equation

We now turn to step two which is solving the transport equation.

Theorem 2.4 (Advective Transport equation)**.** Let $T > 0$ $b \in C^1([0,T] \times \mathbb{R}^d; \mathbb{R}^d)$ bounded, X the flow of b. If $u_0 \in C^0(\mathbb{R}^d; \mathbb{R})$ then there exists $u \in C^1([0,T] \times \mathbb{R}^d; \mathbb{R})$ a solution of

$$\begin{cases} \Big(\partial_t + b(t,x,) \cdot \nabla_x\Big) u(t,x) = 0 \\ u(0,x) = u_0(x) \end{cases} \quad \text{for any } t \in [0,T], x \in \mathbb{R}^d.$$

Moreover the solution is given by

$$u(t,x) = u_0(X(0,t,x)). \tag{2.1}$$

Proof. We start by proving the existence: From theorem 2.3, (1) it follows that the flow $X = X(s,t,x)$ satisfies

$$\frac{\partial}{\partial t_2} X(t_3, t_2, X(t_2, t_1, x)) = \frac{\partial}{\partial t_2} X(t_3, t_1, x) = 0$$

from which we get by writing out the derivative on the LHS explicitly with the chain rule that

$$\begin{aligned} 0 &= (\partial_t X)(t_3, t_2, X(t_2, t_1, x)) + (\nabla_x X)(t_3, t_2, X(t_2, t_1, x) \cdot (\partial_s X)(t_2, t_1, x)) \\ &= (\partial_t X)(t_3, t_2, X(t_2, t_1, x)) + (\nabla_x X)(t_3, t_2, X(t_2, t_1, x) \cdot b(t_2, X(t_2, t_1, x)). \end{aligned}$$

Setting $t_2 = t_1 = t$ and $t_3 = s$ yields

$$\begin{aligned} 0 &= \partial_t X(s,t,x) + \nabla_x X(s,t,x) \cdot b(t,x) \\ &= \Big(\partial_t + b(t,x) \cdot \nabla_x\Big) X(s,t,x) \end{aligned} \tag{2.2}$$

As u_0 is continuously differentiable by assumption,

$$\begin{aligned} \frac{\partial}{\partial t} u_0(X(0,t,x)) &= (\partial_t X) \cdot (\nabla u_0)(X(0,t,x)) \\ &\overset{(2.2)}{=} -\Big(b(t,x) \cdot \nabla_x\Big) X(0,t,x) \cdot (\nabla u_0)(X(0,t,x)) \\ &\overset{\substack{\text{Chain-}\\ \text{rule}}}{=} -b(t,x) \cdot \nabla_x u_0(X(0,t,x)) = -b(t,x) \cdot \nabla_x u(t,x) \end{aligned}$$

which proves that that there exists indeed a solution. We continue with the uniqueness: Take $u = u(t,x)$ a C^1 solution to the transport equation. Then

$$\frac{\partial}{\partial s} u(s, X(s,t,x)) = (\partial_t u)(s, (X(x,t,x)) + \underbrace{\partial_s X(s,t,x)}_{=b(s,(X(s,t,x))} \cdot (\nabla_x u)(s, X(s,t,x)) = 0$$

thus $u(s, X(s,t,x))$ is independent of s. Therefore

$$u(s, X(s,t,x)) = u_0(X(0,t,x)) \quad \text{for any } t, x.$$

Letting $s = t$ (recall $X(t,t,x) = x$) we obtain that the unique solution to the transport equation is given by (2.1), thus proving the theorem. $\square$

Remark 2.5. The *advective form* of the transport equation is given by

$$\Big(\partial_t + b(t,x) \cdot \nabla_x\Big) u(t,x) = 0$$

and the *conservative form* by

$$\partial_t u(t,x) + \underbrace{\operatorname{div}_x \Big(b(t,x) u(t,x)\Big)}_{=(\operatorname{div}_x b)(t,x) u(t,x) + b(t,x) \cdot \nabla_x u(t,x)} = 0.$$

These two expressions coincide for divergence free vector fields b. $\triangle$

Theorem 2.6 (Conservative transport equation)**.** Let $b \in C^2([0,T] \times \mathbb{R}^d; \mathbb{R}^d)$ be a bounded vector field. For $u_0 \in C^1(\mathbb{R}^d)$ there exists a unique $u \in C^1([0,T] \times \mathbb{R}^d)$ solving

$$\begin{cases} \partial_t u + \operatorname{div}_x(bu) = 0 \\ u(t,x)\Big|_{t=0} = u_0(x) \end{cases} \tag{2.3}$$

which is given by $u(t,x) = u_0(X(0,t,x))J(0,t,x)$.

Proof. We proof the uniqueness: Let $u = u(t,x)$ be a solution to (2.3) and let $X = X(s,t,x)$ be the flow of the vector field b. Then

$$\begin{aligned}
\frac{\partial}{\partial s} u(s, X(s,t,x)) = {} & \underbrace{(\partial_t u)(s, X(s,t,x))}_{=-\left(\operatorname{div}_x(bu)\right)(s,X(s,t,x))} + (\nabla_x u)(s, X(s,t,x)) \cdot b(s, X(s,t,x)) \\
&= -\operatorname{div}_x\left(b(t,x)u(t,x)\right) + \underbrace{(\nabla_x u)(s, X(s,t,x)) \cdot b(s, X(s,t,x))}_{=(\operatorname{div}_x(bu))(s,X(s,t,x)) - u(s,X(s,t,x)(\operatorname{div}_x b)(s,X(s,t,x))} \\
&= -(\operatorname{div}_x b)(s, X(s,t,x)) \cdot u(s, X(s,t,x)) \\
&\overset{\text{Thm 2.3}}{=} -\frac{\partial_s J(s,t,x)}{J(s,t,x)} u(s, X(s,t,x)).
\end{aligned}$$

This implies by the product rule for derivatives that

$$\frac{\partial}{\partial s}\Big(u(s, X(s,t,x)) \cdot J(s,t,x)\Big) = 0 \quad \text{i.e.,} \qquad u(t,x) = u_0(X(0,t,x))J(0,t,x)$$

where we used that $J(t,t,s) = 1$ which gives us the uniqueness. To prove the existence recall that

$$\frac{\partial}{\partial t} X(s,t,x) + (\nabla_x X)(s,t,x) \cdot b(t,x) = 0$$

which we saw in a previous theorem. Moreover it is left as an exercise to the reader to prove that

$$\partial_t J(0,t,x) = -\sum_{i,j=1}^{d} \nabla_x^i \Big(b(t,x)J(0,t,x)\Big).$$

Now we see that

$$\begin{aligned}
\partial_t u(t,x) &= u_0(X(t,t,x))\partial_t J(0,t,x)) + J(0,t,x)\sum_{i=1}^{d} (\nabla_x^i)(X(0,t,s))\partial_t X^i(0,t,x) \\
&= -\sum_{i=1}^{d} u_0(X(0,t,x))\nabla_x^i(b^i(t,x)J(0,t,x)) \\
&\qquad + J(0,t,x)\sum_{i=1}^{d} (\nabla_x^i)(X(0,t,s))\partial_t X^i(0,t,x) \\
&= \operatorname{div}(b(t,x))\underbrace{u_0(X(0,t,x))J(0,t,x)}_{=u(t,x)} + \sum_{i=1}^{d} b^i(t,x)J(0,t,x)\nabla_x^i(u_0(X(0,t,x)) \\
&\qquad - \sum_{i,j=1}^{d} J(0,t,x)(\nabla_x^i u_0)(X(t0,t,x))b^i(t,x)\nabla_x^j X^i(0,t,x) \\
&= -\operatorname{div}_x(b(t,x)u(t,x))
\end{aligned}$$

This proves the theorem. $\square$

3. Relaxing the Conditions on the Vector Field

3.1. Repetition: Measure Theory Basics

We saw the existence of classical to the transport equation. Next we want to relax the conditions on the initial datum. in order to do so we will consider solutions that are measures.

First some repetitions:

Definition 3.1. Let Ω be a set. We call $\Sigma \subseteq \mathcal{P}(\Omega)$ a σ-Algebra if the following conditions are satisfied:

(σ1) $\emptyset \in \Sigma$

(σ2) For any $A \in \Sigma$, $A^{\complement} := \Omega \setminus A \in \Sigma$

(σ3) For any sequence $(A_i)_{i\in\mathbb{N}}$ with $A_i \in \Sigma$ for any $i \in \mathbb{N}$, it holds that $\bigcup_{i\in\mathbb{N}} A_i \in \Sigma$.

We call the pair (Ω, Σ) measurable space.

Definition 3.2. Let (Ω, Σ) be a measurable space. A function $\mu : \Sigma \to \mathbb{R}_{\geq 0} \cup \{+\infty\}$ is called a (positive) measure if the following conditions are satisfied:

(m1) $\mu(\emptyset) = 0$ and

(m2) for a sequence $(A_i)_{i\in\mathbb{N}}$ of sets in Σ we have

$$\mu\left(\bigcup_{i\in\mathbb{N}} A_i\right) = \sum_{i\in\mathbb{N}} \mu(A_i).$$

We call this property **σ-additivity**.

The triple (Ω, Σ, μ) is called a **measure space**.

Remark 3.3. This definition can also be extended to $\mu : \Sigma \to \mathbb{R} \cup \{+\infty, -\infty\}$. In this case we speak of *signed* measures. However some things are more tricky with signed measures so we will deal with postive measures for now. △

Example 3.4. Probably the most important example of a σ-algebra is the one generated by the open set of a toplogical space. We call those *Borel-σ-algebras.* △

Definition 3.5 (measurability)**.** Let $(X, \Sigma_X), (Y, \Sigma_Y)$ be two measurable spaces. We call a function $T : X \to Y$ measurable if for any $B \in \Sigma_Y$,

$$T^{-1}[B] := \{x \in X \mid T(x) \in A\} \in \Sigma_X.$$

For μ a measure on (X, Σ_X), we can define the **push-forward** of the measure under the map f as

$$T_\# \mu(A) := \mu(T^{-1}[A]) \quad \text{for any } A \in \Sigma_Y. \tag{3.1}$$

We define the Lebesgue Integral on a measure space (Ω, Σ, μ) by

$$\int_\Omega f \,\mathrm{d}\mu := \sup \left\{ \sum_{i=1}^N a_i \mu(A_i) \mid N \in \mathbb{N}, a_i \in \mathbb{R}, A_i \in \Sigma, 0 \leq \sum_{i=1}^N a_i \mathbb{1}_{A_i}(x) \leq f(x) \forall x \in \Omega \right\}$$

for any non negative measurable function $f : \Omega \to \mathbb{R}$. Moreover for a general measurable function we set

$$\int_\Omega f \,\mathrm{d}\mu := \int_\Omega \max(0, f) \,\mathrm{d}\mu - \int_\Omega \max(-f, 0) \,\mathrm{d}\mu \,.$$

Moreover it can be proven that for the push-forward measure (3.1) it holds that (if f is integrable),

$$\int_Y f(y) \,\mathrm{d}(T_\# \mu)(y) = \int_X f \circ T(x) \,\mathrm{d}\mu(x) \,.$$

3.2. Measure-valued solutions

Definition 3.6. Let $T > 0$ and $b : [0, T] \times \mathbb{R}^d \to \mathbb{R}^d$ a time dependent vector field. We call $(\mu_t)_{t \in [0,T]}$ a family of Borel measures on $\mathbb{R}^d$ a **measure-valued solution** to

$$\begin{cases} \partial_t + \mathrm{div}_x(b\mu_t) = 0 \\ \mu_t|_{t=0} = \mu_0 \end{cases} \tag{3.2}$$

if

$$\int_0^T \int_{\mathbb{R}^d} \frac{\partial}{\partial t} \varphi(t, x) + \nabla_x \varphi(t, x) \cdot b(t, x) \,\mathrm{d}\mu_t(x) \,\mathrm{d}t = 0.$$

for any $\varphi \in C_c^1((0, T) \times \mathbb{R}^d)$, $\mu_t \in C^0([0, T], \mathbb{R})$ i.e., for any $\varphi \in C_c^0(\mathbb{R}^d)$ the map $t \in [0, T] \mapsto \int_{\mathbb{R}^d} \varphi(x) \,\mathrm{d}\mu_t(x) \in \mathbb{R}$ is continuous.

If $u \in C^1([0, T] \times \mathbb{R}^d)$ is a solution to

$$\begin{cases} \partial_t u + \mathrm{div}(b(t, x)u) = 0 \\ u(t, x)|_{t=0} = u_0 \geq 0 \end{cases}$$

then we can define $\nu_t(A) := \int_A u(t, x) \,\mathrm{d}x$ for any $A \in \mathcal{B}(\mathbb{R}^d)$ and see after a calculation that $(\nu_t)_{t \in [0,T]}$ is indeed a measure valued solution to (3.2).

Theorem 3.7. Let $b \in C^1([0, T] \times \mathbb{R}^d; \mathbb{R}^d)$ be a bounded vector field and let μ_0 be a Radon measure on $\mathbb{R}^d$. Then there exists a unique solution to

$$\begin{cases} \partial_t \mu_t + \operatorname{div}(b\mu_t) = 0 \\ \mu_t|_{t=0} = \mu_0 \end{cases}$$

In particular $\mu_t = X(t,0,\cdot)_\# \mu_0$.

Proof. We start with the *existence*: Let $\varphi \in C^1([0,T] \times \mathbb{R}^d)$ and let $X(t,0,x)$ be the flow associated to b. Then

$$\varphi(T, X(T,0,x)) = \varphi(0, X(0,0,x)) = 0 \quad \text{for any } x \in \mathbb{R}^d$$

because of the boundary conditions. By theorem 2.3 the map $t \mapsto \varphi(t, X(t,0,x))$ is continuously differentiable. We thus obtain

$$\begin{aligned} 0 &= \int_{\mathbb{R}^d} \varphi(T, X(T,0,x))\,\mathrm{d}\mu_0(x) - \int_{\mathbb{R}^d} \varphi(0, X(0,0,x))\,\mathrm{d}\mu_0(x) \\ &= \int_{\mathbb{R}^d} \left(\int_0^T \frac{\mathrm{d}}{\mathrm{d}t} \varphi(t, X(t,0,x)) \right) \mathrm{d}t\,\mathrm{d}\mu_0(x) \\ &\overset{\text{Fubini}}{=} \int_0^T \int_{\mathbb{R}^d} \frac{\mathrm{d}}{\mathrm{d}t} \varphi(t, X(t,0,x))\,\mathrm{d}\mu_0(x)\,\mathrm{d}t \\ &= \int_0^T \int_{\mathbb{R}^d} \partial_t \varphi(t, X(t,0,x))\, \underbrace{\partial_t X(t,0,x)}_{=b(t,X(t,0,x))} (\nabla_x \varphi)(t, X(t,0,x))\,\mathrm{d}\mu_0(x)\,\mathrm{d}t \\ &= \int_0^T \int_{\mathbb{R}^d} \varphi(t,x) + b(t,x) \cdot \nabla_x \varphi(t,x)\,\mathrm{d}\mu_t(x)\,\mathrm{d}t \end{aligned}$$

so $\mu_t = X(t,0,\cdot)_\# \mu_0$ is indeed a solution.

For the *uniqueness* part consider $\psi \in C_c^1(\mathbb{R}^d), \chi \in C_c^\infty((0,T))$ and define the function $\phi(t,x) := \chi(t)\psi(X(0,t,x))$. Then by theorem 2.3 $\phi \in C^1((0,T) \times \mathbb{R}^d)$. Thanks to the boundedness of b we obtain

$$\begin{aligned} |X(s,t,y) - y| = |X(s,t,y) - X(t,t,y)| &= \left| \int_t^s \dot{X}(u,t,y)\,\mathrm{d}u \right| \\ &= \left| \int_t^s b(u, X(u,t,y))\,\mathrm{d}u \right| \leq C|t-s| \end{aligned}$$

for any t, s with some constant C only depending on the vectorfield b. Since $\psi \in C_c^1(\mathbb{R}^d)$ there exists a positive radius $R > 0$ such that $\operatorname{spt}(\psi) \subseteq B_R(0)$. This implies that

$$\operatorname{spt} \psi(X(0,t,\cdot)) \subseteq B_{R+Ct}(0) \quad \text{and thus} \quad \psi(X(0,t,\cdot)) \in C_c^1(\mathbb{R}^d)$$

which gives us that $\phi \in C_c^1((0,T) \times \mathbb{R}^d)$ i.e., ϕ is compactly supported. Note that by theorem 2.4 $\psi \circ X$ satisfies the transport equation

$$(\partial_t + b(t,x) \cdot \nabla_x)\psi(X(0,t,x)) = 0$$

which implies

$$(\partial_t + b(t,x) \cdot \nabla_x)\phi(t,x) = \dot{\chi}(t)\psi(X(0,t,x)) \tag{3.3}$$

for any $(t,x) \in (0,T) \times \mathbb{R}^d$.

Now let $(\mu_t)_{t\in[0,T]}$ be a measure-valued solution of the transport equation and let $\nu_t := X(0,t,\cdot)_{\#}\mu_t$. Then

$$\begin{aligned}\int_0^T \int_{\mathbb{R}^d} \left(\psi(x)\frac{\mathrm{d}}{\mathrm{d}t}\chi(t)\right) \mathrm{d}\nu_t(x)\,\mathrm{d}t &= \int_0^T \int_{\mathbb{R}^d} \dot\chi(t)\cdot\psi(X(0,t,x))\,\mathrm{d}\mu_t\,\mathrm{d}t \\ &\overset{(3.3)}{=} \int_0^T \int_{\mathbb{R}^d} (\partial_t + b(t,x)\cdot\nabla_x)\phi(t,x)\,\mathrm{d}\mu_t(x)\,\mathrm{d}t = 0\end{aligned} \tag{3.4}$$

Note that

$$w(t) := \int_{\mathbb{R}^d} \psi(x)\,\mathrm{d}\nu_t(x) = \int_{\mathbb{R}^d} \psi\circ X(0,t,x)\,\mathrm{d}\mu_t(x)$$

is a continuous function and by (3.4)

$$\int_0^T \left(\frac{\mathrm{d}}{\mathrm{d}t}\xi(t)\right) w(t)\,\mathrm{d}t = 0 \quad \text{for any } \xi \in C_c^\infty((0,T)).$$

Now, choose $\eta \in C_c^\infty((0,T))$ such that $\int_0^T \eta(t)\,\mathrm{d}t = 1$. We may represent an arbitary $\chi \in C_c^\infty((0,T))$ as $\chi = A\eta + \dot\xi$ where $A \in \mathbb{R}$ and $\xi \in C_c^\infty((0,T))$ are given by

$$A = \int_0^T \chi(t)\,\mathrm{d}t \quad \text{and} \quad \xi(t) = \int_0^t \chi(u) - A\eta(u)\,\mathrm{d}u\,.$$

Then:

$$\int_0^T \chi(t)w(t)\,\mathrm{d}t = A\underbrace{\int_0^T \eta(t)w(t)\,\mathrm{d}t}_{=:C} + \underbrace{\int_0^T \dot\xi(t)w(t)\,\mathrm{d}t}_{=0}$$

implying

$$\int_0^T \chi(t)(w(t)-C)\,\mathrm{d}t = 0 \quad \text{for any } \chi \in C_c^\infty((0,T))$$

which gives us that $w(t) = C$ for almost any t . Thanks to the continuity we can conclude this fact for any t i.e., $w \equiv C$. We thus obtain

$$\int_{\mathbb{R}^d} \psi(x)\,\mathrm{d}\nu_t(x) = \int_{\mathbb{R}^d} \psi(x)\,\mathrm{d}\nu_0(x) = \int_{\mathbb{R}^d} \psi(x)\,\mathrm{d}\mu_0(x) \qquad \text{for all } \psi \in C_c^1(\mathbb{R}^d).$$

We thus conclude

$$X(0,t,\cdot)_{\#}\mu_t = \nu_t = \mu_0 \quad \text{i.e.,} \qquad \mu_t = X(t,0,\cdot)_{\#}\mu_0 \quad \text{for all } t \in [0,T].$$

This finishes the proof. □

3.3. Repetition: Weak derivatives and Sobolev spaces

In the following let $\Omega \subseteq \mathbb{R}^d$ be an open and non-empty set. We denote the *support of a function* $\varphi : D \subseteq \mathbb{R}^d \to \mathbb{R}^n$ by

$$\operatorname{spt}\varphi = \overline{\{x \in D \mid \varphi(x) \neq 0\}}.$$

Moreover for a function $\varphi \in C^k(D,\mathbb{R}^n)$ we will use the *multiindex* notation for derivatives. That is, for $\alpha = (\alpha_i)_{i=1}^d \in \mathbb{N}_0^d$ with

$$|\alpha| := \sum_{i=1}^d \alpha_i \leq k$$

we write

$$\partial^\alpha\varphi(x) := \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_d}}{\partial x_d^{\alpha_d}}\varphi(x_1,\ldots,x_d).$$

Definition 3.8. We define the space of test functions $\mathcal{D}(\Omega)$ as set of all smooth functions i.e.,

$$\mathcal{D}(\Omega) := \{\varphi \in C^\infty(\Omega) \mid \varphi \text{ has compact support in } \Omega\} = C_c^\infty(\Omega)$$

and we endow this set by the following notion of convergence: A sequence of functions $(\varphi_n)_{n\in\mathbb{N}} \in (C_c^\infty(\Omega))^{\mathbb{N}}$ converges in $\mathcal{D}(\Omega)$ if and only if there exists $K \subset \Omega$ compact such that $\operatorname{spt}\varphi_n \subseteq K$ for any $n \in \mathbb{N}$ and for any multi-index $\alpha \in \mathbb{N}_0^d$,

$$\sup_{x\in K} |\partial^\alpha \varphi_n(x) - \partial^\alpha \varphi(x)| \xrightarrow{n\to+\infty} 0.$$

Definition 3.9. A **distribution** T is a continuous linear functional $T : \mathcal{D}(\Omega) \to \mathbb{C}$. We will use the notation $T(\varphi) =: \langle T, \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}}$ and we will denote the space of distributions by $\mathcal{D}'(\Omega)$ and we say that a sequence of distributions $(T_j)_{j\in\mathbb{N}} \in (\mathcal{D}'(\Omega))^{\mathbb{N}}$ converges to $T \in \mathcal{D}'(\Omega)$ if we have pointwise convergence i.e.,

$$|\langle T_j, \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} - \langle T, \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}}| \xrightarrow{j\to+\infty} 0 \quad \text{for any } \varphi \in \mathcal{D}(\Omega).$$

Exercise 3.10. (1) Show that for any Borel measure on Ω, $\langle \mu, \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} = \int_\Omega \varphi \,\mathrm{d}\mu$ is a distribution.

(2) Show that for any $f \in L^1_{\text{loc}}(\Omega)$, $\langle f, \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} = \int_\Omega f(x)\varphi(x)\,\mathrm{d}x$ is a distribution.

Definition 3.11. The **distributional derivative** is defined as

$$\langle \partial^\alpha T, \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} := (-1)^{|\alpha|}\langle T, \partial^\alpha\varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} \quad \text{for any } \alpha \in \mathbb{N}_0^d, \varphi \in \mathcal{D}(\Omega).$$

This formula is motivated by the integration by parts formula.

Remark 3.12. Every distribution is infinitely often differentiable but the derivative is not always straight forward to compute. If we for example let $\Omega = (0,1)$ and let f be the Cantor function which is almost everywhere differentiable with derivative zero. Then it holds that

$$-\langle f, \varphi'\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} = -\int_0^1 f(x)\varphi'(x)\,\mathrm{d}x \neq \int_0^1 \underbrace{f'(x)}_{=0}\varphi(x)\,\mathrm{d}x = 0$$

thus the almost everywhere derivative of the Cantor-function and its distributional derivative do not match. However, if we take $f \in C^1((0,1)$ we get by the integration by parts formula that for any $\varphi \in \mathcal{D}(\Omega)$,

$$\langle f', \varphi\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}} = -\int_0^1 f(x)\varphi'(x)\,\mathrm{d}x = \int_0^1 \frac{\mathrm{d}f}{\mathrm{d}x}(x)\varphi(x)\,\mathrm{d}x = \left\langle \frac{\mathrm{d}f}{\mathrm{d}x}, \varphi\right\rangle_{\mathcal{D}'\leftrightarrow\mathcal{D}}$$

where we denoted the distributional derivative with f' and the classical one with $\frac{\mathrm{d}f}{\mathrm{d}x}$. In most cases we use the same notation for the classical and distributional derivative. △

Definition 3.13. Let $p \in [1, +\infty]$ and $k \in \mathbb{N}_0$. We define the **Sobolev-Space**

$$W^{k,p}_{\text{loc}}(\Omega) = \left\{f : \Omega \to \mathbb{C} \mid f \in L^p_{\text{loc}}(\Omega) : \forall\alpha \in \mathbb{N}_0^d \text{ s.t.} |\alpha| \leq k, \partial^\alpha f \in L^p_{\text{loc}}(\Omega)\right\}$$

and

$$W^{k,p}(\Omega) = \Big\{ f : \Omega \to \mathbb{C} \mid f \in L^p(\Omega) : \forall \alpha \in \mathbb{N}_0^d \text{ s.t.} |\alpha| \le k, \partial^\alpha f \in L^p(\Omega) \Big\}.$$

We endow these spaces with the norm

$$\|f\|_{W^{k,p}(\Omega)} := \begin{cases} \Big(\sum\limits_{\substack{\alpha \in \mathbb{N}_0^d \\ |\alpha| \le k}} \int_\Omega |\partial^\alpha f|^p \, \mathrm{d}x \Big)^{1/p} & \text{if } 1 \le p < +\infty \\ \sum\limits_{\substack{\alpha \in \mathbb{N}_0^d \\ |\alpha| \le k}} \operatorname{ess\,sup}_\Omega |\partial^\alpha f| & \text{if } p = +\infty \end{cases}.$$

If $p = 2$ we use the notation $H^k(\Omega) := W^{k,2}(\Omega)$ which is a Hilbert space.

Theorem 3.14. If Ω is a measurable subset of $\mathbb{R}^d$ and $(f_n)_{n \in \mathbb{N}} \in (L^\infty(\Omega))^{\mathbb{N}}$ a bounded sequence there exists a weakly convergent subsequence $(f_{n_j})_{j \in \mathbb{N}}$ i.e., there is $f \in L^\infty(\Omega)$ such that

$$\int_\Omega \varphi(x) f_{n_j}(x) \, \mathrm{d}x \xrightarrow{j \to +\infty} \int_\Omega \varphi(x) f(x) \, \mathrm{d}x \qquad \text{for any } \varphi \in \mathcal{D}(\Omega).$$

Proof. See literature. $\square$

We now study weak solutions to the transport equation.

Definition 3.15. We say that $u : [0,T) \times \mathbb{R}^d \to \mathbb{R}^d$ is a **weak solution** of the transport equation

$$\begin{cases} \partial_t + \operatorname{div}(bu) = 0 \\ u(0,x) = u_0(x) \end{cases}$$

if for any $\phi \in C_c^\infty([0,T) \times \mathbb{R}^d)$ it holds that

$$0 = \int_{\mathbb{R}^d} u_0(x) \phi(0,x) \, \mathrm{d}x + \int_0^T \int_{\mathbb{R}^d} u(t,x) \Big\{ \partial_t \phi(t,x) + b(t,x) \nabla_x \phi(t,x) \Big\} \, \mathrm{d}x \, \mathrm{d}t$$

and a **weak solution** of

$$\begin{cases} \partial_t u + b \cdot \nabla_x u = 0 \\ u(0,x) = u_0(x) \end{cases}$$

if for any $\varphi \in C_c^\infty([0,T) \times \mathbb{R}^d; \mathbb{R}^d)$,

$$0 = \int_{\mathbb{R}^d} u_0(x) \varphi(0,x) \, \mathrm{d}x + \int_0^T \int_{\mathbb{R}^d} u(t,x) \Big\{ \partial_t \varphi(t,x) + \operatorname{div}_x \big(b(t,x) \varphi(t,x) \big) \Big\} \, \mathrm{d}x \, \mathrm{d}t \, .$$

Theorem 3.16. Let $b, \operatorname{div}_x b, u_0 \in L^\infty(\mathbb{R}^d)$ where b is a time independent vector field $b : \mathbb{R}^d \to \mathbb{R}^d$. For any $T > 0$ there exists a weak solution $u \in L^\infty([0,T] \times \mathbb{R}^d)$ of

$$\begin{cases} \partial_t u + \operatorname{div}(bu) = 0 \\ u(0,x) = u_0 \end{cases}.$$

Remark 3.17. The proof consists of the following three steps:

(1) Regularize b and u_0,

(2) Prove the existence of the solution for the regularized system and

(3) Obtain the solution by a limit argument of the regularized version.

△

Proof of Theorem 3.16. Step 1: Let $(\eta_\epsilon)_{\epsilon>0}$ be a family of mollifiers. We then mollify and obtain $b_\epsilon := b \star \eta_\epsilon$ and $u_0^{(\epsilon)} = u_0 \star \eta_\epsilon$. Moreover from the properties of the mollification we have

$$\sup_{\epsilon>0} \|b_\epsilon\|_{L^\infty} \leq C_1 \in \mathbb{R}, \quad \sup_{\epsilon>0} \|\operatorname{div}_x b_\epsilon\|_{L^\infty} \leq C_2 \in \mathbb{R}, \quad \sup_{\epsilon>0} \|u_0^{(\epsilon)}\|_{L^\infty} \leq C_3 \in \mathbb{R}. \tag{3.5}$$

and

$$\lim_{\epsilon\downarrow 0} \|\xi^\epsilon - \xi\|_{L^1_{\text{loc}}} = 0 \quad \text{for all } \xi \in \{b, u_0\}.$$

Since we already solved the transport equation in the classical case we find for any $\epsilon > 0$ a classical solution to

$$\begin{cases} \partial_t u^\epsilon + \operatorname{div}_x(b^\epsilon u^\epsilon) = 0 \\ u^\epsilon(0,x) = u_0^\epsilon(x) \end{cases}.$$

Moreover $u^\epsilon(t,x) = u_0^\epsilon(X^\epsilon(0,t,x))J^\epsilon(0,t,x)$ where X^ϵ is the flow associated to b^ϵ and $J^\epsilon(s,t,x) = \det(\nabla_x X^\epsilon(s,t,x))$. By theorem 2.3,

$$\partial_s J^\epsilon(s,t,x) = \operatorname{div}_x b^\epsilon(X^\epsilon(s,t,x))J^\epsilon(s,t,x)$$

which implies that

$$J^\epsilon(s,t,x) = J^\epsilon(0,t,x) \exp\left\{\int_0^s \operatorname{div}_x b^\epsilon(X^\epsilon(u,t,x))\,\mathrm{d}u\right\} \leq J^\epsilon(0,t,x)\mathrm{e}^{C_2 s}.$$

Setting $s = t$ we obtain that

$$|J^\epsilon(0,t,x)| \leq \underbrace{|J^\epsilon(t,x,x)|}_{=1}\mathrm{e}^{Ct}$$

which implies

$$\|J^\epsilon(0,\cdot,\cdot)\|_{L^\infty([0,T]\times\mathbb{R}^d)} \leq C.$$

Together with (3.5) we obtain that u^ϵ is bounded in L^∞. Recall that

$$\lim_{\epsilon\downarrow 0} \|b^\epsilon - b\|_{L^1_{\text{loc}}} = 0 \quad \text{thus} \quad \lim_{\epsilon\downarrow 0} \|u_0^\epsilon - u_0\|_{L^1_{\text{loc}}} = 0.$$

By the above arguments and theorem 3.14 there exist a sequence $(\epsilon_n)_{n\in\mathbb{N}}$ with $\epsilon_n \to 0$ as $n \to +\infty$ and $u \in L^\infty([0,T]\times\mathbb{R}^d)$ such that $u_{\epsilon_n} \overset{L^\infty}{\rightharpoonup} u$ meaning

$$\int_0^T \int_{\mathbb{R}^d} v(t,x)u_{\epsilon_n}(t,x)\,\mathrm{d}x\,\mathrm{d}t \xrightarrow{n\to+\infty} \int_0^T \int_{\mathbb{R}^d} v(t,x)u(t,x)\,\mathrm{d}x\,\mathrm{d}t \;\text{ for any } v \in L^1([0,T]\times\mathbb{R}^d).$$

Now, let $\phi \in C_c^\infty([0,T)\times\mathbb{R}^d)$ and note that

$$0 = \int_{\mathbb{R}^d} u_0^{\epsilon_n}(x)\phi(0,x) + \int_0^T \int_{\mathbb{R}^d} u_{\epsilon_n}(t,x)\left\{\partial_t\phi(t,x) + b^{\epsilon_n}(x)\cdot\nabla_x\phi(t,x)\right\}\mathrm{d}x\,\mathrm{d}t \tag{3.6}$$

which implies

$$\begin{aligned}
&\int_{\mathbb{R}^d} u_0(x)\phi(0,x)\,\mathrm{d}x + \int_0^T \int_{\mathbb{R}^d} u(t,x)\Big\{\partial_t\phi(t,x) + b(x)\cdot\nabla_x\phi(t,x)\Big\}\,\mathrm{d}x\,\mathrm{d}t \\
&\quad \overset{(3.6)}{=} \int_{\mathbb{R}^d} \Big\{u_0(x) - u_0^{\epsilon_n}(x)\Big\}\phi(0,x)\,\mathrm{d}x \\
&\quad + \int_0^T \int_{\mathbb{R}^d} \Big(u(t,x) - u_{\epsilon_n}(t,x)\Big)\Big(\partial_t\phi(t,x) + b(x)\nabla_x\phi(t,x)\Big)\,\mathrm{d}x\,\mathrm{d}t \\
&\qquad + \int_0^T \int_{\mathbb{R}^d} u_{\epsilon_n}(t,x)\big(b(x) - b^{\epsilon_n}(x)\big)\cdot\nabla_x\phi(t,x)\,\mathrm{d}x\,\mathrm{d}t \\
&\quad =: \mathrm{I}_\epsilon + \mathrm{II}_\epsilon + \mathrm{III}_\epsilon
\end{aligned}$$

We are now going to estimate those terms peu à peu:

$$\mathrm{I}_\epsilon \le \|\phi(0,\cdot)\|_{L^\infty(\mathbb{R}^d)}\|u_0 - u_0^{\epsilon_n}\|_{L^1_{\mathrm{loc}}(\mathbb{R}^d)} \xrightarrow{n\to+\infty} 0.$$

To estimate II_ϵ we use that ϕ has compact support to find a compact set $K \subset [0,T]\times\mathbb{R}^d$ such that

$$\operatorname{spt}\Big(\partial_t\phi(t,x) + b(x)\cdot\nabla_x\phi(t,x)\Big) \subseteq K.$$

Then:

$$\begin{aligned}
\mathrm{II}_\epsilon &\le \iint_K |u - u_{\epsilon_n}|\cdot\big|\partial_t\phi(t,x) + b(x)\cdot\nabla_x\phi(t,x)\big|\,\mathrm{d}x\,\mathrm{d}t \\
&\le \|\partial_t\phi\|_{L^\infty([0,T]\times\mathbb{R}^d)} + \|b\|_{L^\infty(\mathbb{R}^d)}\|\nabla_x\phi\|_{L^\infty([0,T]\times\mathbb{R}^d)} \underbrace{\iint_K |u - u_{\epsilon_n}|\,\mathrm{d}x\,\mathrm{d}t}_{\|u-u_{\epsilon_n}\|_{L^1_{\mathrm{loc}}([0,T]\times\mathbb{R}^d)}} \xrightarrow{n\to+\infty} 0.
\end{aligned}$$

By a similar argument,

$$\mathrm{III}_\epsilon \le T\|u_{\epsilon_n}\|_{L^\infty([0,T]\times\mathbb{R}^d)}\|\nabla_x\phi\|_{L^\infty([0,T]\times\mathbb{R}^d)}\|b - b^{\epsilon_n}\|_{L^1_{\mathrm{loc}}(\mathbb{R}^d)} \xrightarrow{n\to+\infty} 0.$$

This shows the result. □

Remark 3.18. This result an be generalized to time dependent vector fields b and to the other transport equation. △

Example 3.19. If we loosen the assumption for the vector field b further, we run into danger of losing uniqueness of the solution to the equation. Consider the case $d = 1$ and let $u_0 \in C^1(\mathbb{R}), b(x) = \operatorname{sign}(x)$ where

$$\operatorname{sign}(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0 \end{cases}.$$

Then for any $h \in C^1(\mathbb{R})$ with $h(0) = u_0(0)$ we obtain that

$$u(t,x) = \begin{cases} u_0(x+t), & x + t < 0 \\ u_0(x-t), & x - t > 0 \\ h(t - |x|), & |x| \le t \end{cases}$$

is a weak solution of the transport equation which reads in this case as

$$\begin{cases} \partial_t u + b\dfrac{\mathrm{d}u}{\mathrm{d}x} = 0 \\ u(0,x) = u_0(x) \end{cases}.$$

This shows that the solution is no longer unique if we relax the conditions on the vector field b. △

3.4. Renormalized Solutions

To solve the uniqueness problem of the transport equation we introduce the notion of renormalized solutions. To understand this better, let $\beta \in C^1(\mathbb{R})$ and $u \in C^1([0,T]\times\mathbb{R}^d)$ satisfying

$$\partial_t u(t,x) + b(t,x)\cdot\nabla_x u(t,x) = 0.$$

Multiplying both sides by $\beta'(u(t,x))$ yields

$$0 = \partial_t \beta(u(t,x)) + b(t,x)\nabla_x \beta(u(t,x)) \tag{3.7}$$

The idea of a renormalized solution is to relax the regularity for b and u but require that (3.7) holds to infer uniqueness of solutions.

Definition 3.20. Let $b : [0,T]\times\mathbb{R}^d \to \mathbb{R}^d$ such that $\operatorname{div}_x b \in L^\infty([0,T];L^\infty(\mathbb{R}^d)$ and let $u \in L^\infty([0,T]\times\mathbb{R}^d)$ be a solution of

$$\partial_t u + b\cdot\nabla_x u = 0$$

We say that u is a **renormalized solution** if for any $\beta \in C^1(\mathbb{R})$,

$$\frac{\partial}{\partial t}(\beta\circ u) + b\cdot\nabla_x(\beta\circ u) = 0 \quad \text{in } \mathcal{D}'([0,T]\times\mathbb{R}^d).$$

The notion of renormalized solutions can be used to formulate a condition on the vector field.

Definition 3.21. Assume that $b, \operatorname{div}_x b \in L^1([0,T];L^\infty(\mathbb{R}^d)$. Then b is said to have the **renormalization property** if every L^∞ solution of

$$\partial_t u + b\cdot\nabla_x u = 0$$

is a renormalized solution

Theorem 3.22. Let $b : [0,T]\times\mathbb{R}^d \to \mathbb{R}^d$ be such that $b \in L^\infty([0,T]\times\mathbb{R}^d)$, $\operatorname{div}_x b \in L^1([0,T];L^\infty(\mathbb{R}^d))$. We define $\bar{b} : (-\infty,T]\times\mathbb{R}^d$ by

$$\bar{b}(t,x) = \begin{cases} b(t,x) & 0\le t\le T \\ 0 & t<0 \end{cases}.$$

If b has the renormalization property then L^∞-solutions to

$$\partial_t u + b \cdot \nabla_x u = 0$$

are unique.

Proof. Let $u(0,\cdot) \equiv 0$. Extending b to negative times with $\bar{b}$ we obtain that

$$\begin{cases} \partial_t u + \bar{b} \cdot \nabla_x u = 0 \\ u(t,0) = 0 \quad \text{ for any } t \le 0 \end{cases} \quad \text{in } \mathcal{D}'((-\infty, T] \times \mathbb{R}^d)$$

Renormalizing with $\beta(x) := x^2$ gives

$$\begin{cases} \partial_t u^2 + \bar{b} \cdot \nabla_x u^2 = 0 \\ u(t,\cdot) = 0 \text{ for any } t \le 0 \end{cases} \quad \text{in } \mathcal{D}'((-\infty, T] \times \mathbb{R}^d).$$

Fix $R > 0, \eta > 0$. Consider $\varphi \in C_c^\infty((-\infty, \mathbb{R}^d)$ such that $\varphi \ge 0$ and $\varphi|_{[0,T-\eta]\times B_R(0)} \equiv 0$ and

$$\partial_t \varphi(t,x) \le \|b\|_{L^\infty} |\nabla_x \varphi(t,x)| \quad \text{ on } [0,T) \times \mathbb{R}^d.$$

Define $f(t) := \int_{\mathbb{R}^d} u^2(t,x)\varphi(t,x)\,\mathrm{d}x$. For $\psi \in C_c^\infty((-\infty,T))$ with $\psi \ge 0$ we obtain (because we loose all the boundary terms on the integral because we excluded $-\infty$ and T),

$$\begin{aligned} 0 &= \int_{-\infty}^T \int_{\mathbb{R}^d} u^2(t,x) \Big\{ \partial_t \varphi(t,x) + \psi(t) + \operatorname{div}\big(\bar{b}(t,x)\varphi(t,x)\psi(t)\big) \Big\}\,\mathrm{d}x\,\mathrm{d}t \\ &= \int_{-\infty}^T f(t) \frac{\mathrm{d}}{\mathrm{d}t}\psi(t)\,\mathrm{d}t + \int_{-\infty}^T \int_{\mathbb{R}^d} u^2(t,x)\psi(t) \Big\{ \partial_t \varphi(t,x) + \bar{b}(t,x) \cdot \nabla_x \varphi(t,x) \Big\}\,\mathrm{d}x\,\mathrm{d}t \\ &\qquad + \int_{-\infty}^T \int_{\mathbb{R}^d} u^2(t,x)\varphi(t,x)\psi(t) \operatorname{div}_x \bar{b}(t,x)\,\mathrm{d}x\,\mathrm{d}t \\ &\le \int_{-\infty}^T f(t) \frac{\mathrm{d}}{\mathrm{d}t}\psi(t)\,\mathrm{d}t + \int_{-\infty}^T \|\operatorname{div}_x \bar{b}(t,\cdot)\|_{L^\infty(\mathbb{R}^d)} \psi(t) f(t)\,\mathrm{d}t\,, \end{aligned}$$

which implies

$$\int_{-\infty}^T f(t) \frac{\mathrm{d}}{\mathrm{d}t}\psi(t)\,\mathrm{d}t \le \int_{-\infty}^T f(t) \|\operatorname{div}_x \bar{b}(t,\cdot)\|_{L^\infty(\mathbb{R}^d)} f(t)\,\mathrm{d}t\,.$$

This leads to

$$\frac{\mathrm{d}}{\mathrm{d}t} f(t) \le \|\operatorname{div}_x \bar{b}(t,\cdot)\|_{L^\infty(\mathbb{R}^d)} f(t) \quad \text{for almost any } t \in (-\infty, T]$$

and using Grönwall's inequality,

$$\frac{\mathrm{d}}{\mathrm{d}t} f(t) \exp\left\{ - \int_0^T \|\operatorname{div}_x \bar{b}(s,\cdot)\|_{L^\infty(\mathbb{R}^d)}\,\mathrm{d}s \right\} \le 0 = f(0),$$

thus $f(t) = 0$ for any $t \in (-\infty, T]$. Since η and R are chosen arbitrary we can conclude that $u(t,\cdot) \equiv 0$ for any $t \in [0,T]$ by letting $\eta, R \to 0$. $\square$

We saw that renormalization property ensures the uniqueness of the transport equation. Di Perna and Lions proved that Sobolev-regularity implies the renormalization property:

Theorem 3.23. Let b be a bounded vector field, $b \in L^1_{\mathrm{loc}}(I, W^{1,1}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^d))$ where $I \subset \mathbb{R}$ is an intervall. Then b has the renormalization property.

Proof. The proof of this theorem can be found in the literature. $\square$

4. Wasserstein Distance

In the following chapters we are interested in the well-posedness and derivation of the Vlasov equation. To do this we have to introduce a specific distance between two probability measures which we will call the *Wasserstein distance*.

First a reminder about Lipschitz-continuous functions:

Definition 4.1. Let $\varphi \in C^0(\mathbb{R}^d, \mathbb{R}^m)$. We say that φ is **Lipschitz continuous** if the so-called **Lipschitz-Norm**

$$\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^d)} := \sup_{\substack{x \neq y \\ x,y \in \mathbb{R}^d}} \frac{|\varphi(x) - \varphi(y)|}{|x - y|} < +\infty$$

is finite. We will denote the collection of Lipschitz continuous functions $\mathbb{R}^d \to \mathbb{R}^m$ by $\mathrm{Lip}(\mathbb{R}^d, \mathbb{R}^m)$.

Remark 4.2. The Lipschitz-norm is not a norm but a semi-norm. This is due to the fact that if φ is constant, $\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^d)} = 0$. This however induces a norm on the quotient space $\mathrm{Lip}(\mathbb{R}^d)/\sim$ where $\sim$ is the equivalence relation given by

$$f \sim g \quad \text{if} \quad f - g \text{ is constant.}$$

Said quotient space is also a Banach space. △

Definition 4.3. We denote by $\mathcal{P}_1(\mathbb{R}^d)$ the space of all probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ with finite first moment i.e., $\mu \in \mathcal{P}_1(\mathbb{R}^d)$ if and only if μ is a positive Borel measure on $\mathbb{R}^d$ with

$$\mu(\mathbb{R}^d) = 1 \quad \text{and} \quad \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x) < +\infty.$$

The **Wasserstein distance** (also called Monge/Kantorovich/Rubenstein distance) of order one is given by

$$\mathcal{W}_1(\mu, \nu) = \sup \left\{ \int_{\mathbb{R}^d} \varphi \,\mathrm{d}\mu - \int_{\mathbb{R}^d} \varphi \,\mathrm{d}\nu \mid \varphi \in \mathrm{Lip}(\mathbb{R}^d), \|\varphi\|_{\mathrm{Lip}(\mathbb{R}^d)} \leq 1 \right\}, \tag{4.1}$$

where $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^d)$.

Remark 4.4. (4.1) is always non-negative. Indeed, this holds because for $\varphi \in \mathrm{Lip}(\mathbb{R}^d)$, $-\varphi \in \mathrm{Lip}(\mathbb{R}^d)$ where $-\varphi$ and φ have the same Lipschitz norm. △

Remark 4.5. The Wasserstein distance of order one is usually defined in its dual formulation which is given by

$$\mathcal{W}_1(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^d} |x - y| \,\mathrm{d}\pi(x, y)$$

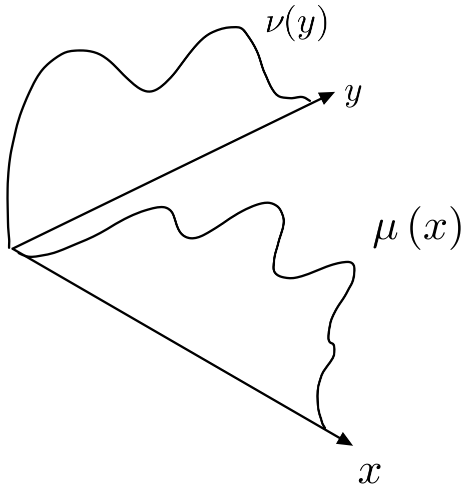

Figure 4.1.: One can think of the Wasserstein distance as a minimal cost transport plan. Imagine we are transporting mass with distribution $\mu(x)$ to another place and transform it into the distribution $\nu(y)$. This only makes sense if both distributions have the same total mass. We thus think think of $\pi \in \Pi(\nu, \mu)$ as a transport plan where $\mathrm{d}\pi(x, y)$ describes how much mass from point x to y is moved. Moreover, we think of $|x - y|$ as the cost function of moving said mass from x to y. $\mathcal{W}_1(\nu, \mu)$ describes the minimal cost that is needed to change the mass distribution from ν to μ.

where $\Pi(\mu, \nu)$ is the set of measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν i.e., every $\pi \in \Pi(\mu, \nu)$ satisfies

$$\pi(A \times \mathbb{R}^d) = \mu(A) \quad \text{and} \quad \pi(\mathbb{R}^d \times A) = \nu(A) \quad \text{for any } A \in \mathcal{B}(\mathbb{R}^d).$$

Analogously we could define the Wasserstein distance of order p by

$$\mathcal{W}_p(\mu, \nu) := \inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^d} |x - y|^p \, \mathrm{d}\pi(x, y) .$$

For references see C. Villani Topics in optimal transport and Figure 4.1. △

Lemma 4.6. $\mathcal{W}_1$ is indeed a distance/metric on $\mathcal{P}_1(\mathbb{R}^d)$.

Proof. First we prove that the Wasserstein distance is finite for any probability measures $\nu, \mu \in \mathcal{P}_1(\mathbb{R}^d)$: Let $\varphi \in \mathrm{Lip}(\mathbb{R}^d)$ such that $\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^d)} \leq 1$. Then, using that μ and ν are probability measures we obtain

$$\begin{aligned}
\left| \int_{\mathbb{R}^d} \varphi \, \mathrm{d}\mu - \int_{\mathbb{R}^d} \varphi \, \mathrm{d}\nu \right| &= \left| \int_{\mathbb{R}^d} \Big(\varphi(x) - \varphi(0) \Big) \, \mathrm{d}\mu(x) - \int_{\mathbb{R}^d} \Big(\varphi(x) - \varphi(0) \Big) \, \mathrm{d}\nu(x) \right| \\
&\leq \int_{\mathbb{R}^d} |\varphi(x) - \varphi(0)| \, \mathrm{d}\mu(x) + \int_{\mathbb{R}^d} |\varphi(x) - \varphi(0)| \, \mathrm{d}\nu(x) \\
&\leq \int_{\mathbb{R}^d} |x| \, \mathrm{d}\mu(x) + \int_{\mathbb{R}^d} |x| \, \mathrm{d}\nu(x) < +\infty
\end{aligned}$$

we thus found a uniform bound for any Lipschitz function implying $\mathcal{W}_1(\mu, \nu) < +\infty$ for any $\mu, \nu \in \mathcal{P}_1(\mathbb{R}^6)$. It is obvious from the definition that $\mathcal{W}_1(\nu, \mu) = \mathcal{W}_1(\mu, \nu)$ for any μ, ν. Moreover it is easy to see that the triangle inequality holds.

We thus are left to prove that $\mathcal{W}_1(\nu,\mu) = 0$ implies that $\nu \equiv \mu$. Let $\varphi \in \mathrm{Lip}(\mathbb{R}^d)$ with $\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^d)} \leq 1$. If $\mathcal{W}_1(\nu,\mu) = 0$ it holds that

$$\int_{\mathbb{R}^d} \varphi \,\mathrm{d}\mu = \int_{\mathbb{R}} \varphi \,\mathrm{d}\nu \,.$$

We now pick a set $E \in \mathcal{B}(\mathbb{R}^d)$, a function $\psi \in \mathrm{Lip}(\mathbb{R}^d)$, $\|\psi\|_{\mathrm{Lip}(\mathbb{R}^d)} \leq 1$ and define two functions

$$\psi_M(x) := M^d \psi(Mx), \quad \chi_{E,M} := \mathbb{1}_E \star \psi_M.$$

Then $\chi_{E,M}$ converges to $\mathbb{1}_E$ almost everywhere on $\mathbb{R}^d$ and $\chi_{E,M}$ is a Lipschitz function with Lipschitz norm smaller or equal to one. We thus obtain by the dominated convergence theorem,

$$\mu(E) = \lim_{M\to+\infty} \int_{\mathbb{R}^d} \chi_{E,M} \,\mathrm{d}\mu = \lim_{M\to+\infty} \int_{\mathbb{R}^d} \chi_{E,M} \,\mathrm{d}\nu = \nu(E)$$

and conclude by the arbitrariness of E that $\nu \equiv \mu$. □

Definition 4.7. Let $\mu_k, \mu \in \mathcal{P}(\mathbb{R}^d)$ for any $k \in \mathbb{N}$. We say that μ_k converges weakly to μ, written as $\mu_k \rightharpoonup \mu$ if

$$\lim_{k\to+\infty} \int_{\mathbb{R}^d} \varphi \,\mathrm{d}\mu_k = \int_{\mathbb{R}^d} \varphi \,\mathrm{d}\mu \quad \text{for any } \varphi \in C_b^0(\mathbb{R}^d).$$

Lemma 4.8. Let $\mu_k, \mu \in \mathcal{P}_1(\mathbb{R}^d)$ for any $k \in \mathbb{N}$. The following statements are equivalent:

(1) $\mu_k \rightharpoonup \mu$ and $\displaystyle\int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x) \xrightarrow{k\to+\infty} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x)\,.$

(2) $\mu_k \rightharpoonup \mu$ and $\displaystyle\limsup_{k\to+\infty} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x) \leq \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x)\,.$

(3) $\mu_k \rightharpoonup \mu$ and $\displaystyle\lim_{R\to+\infty} \limsup_{k\to+\infty} \int_{|x|\geq R} |x| \,\mathrm{d}\mu_k(x) = 0.$

(4) $\mu_k \rightharpoonup \mu$ and for any $\varphi \in C^0(\mathbb{R}^d)$ such that there exists $C > 0$ with $|\varphi(x)| \leq C(1+|x|)$ for any $x \in \mathbb{R}^d$,

$$\int_{\mathbb{R}^d} \varphi \,\mathrm{d}\mu_k \xrightarrow{k\to+\infty} \int_{\mathbb{R}^d} \varphi \,\mathrm{d}\mu\,.$$

Definition 4.9. Let $\mu_k, \mu \in \mathcal{P}_1(\mathbb{R}^d)$ for any $k \in \mathbb{N}$. We say that μ_k converges weakly to μ in $\mathcal{P}_1(\mathbb{R}^d)$, written as $\mu_k \rightharpoonup \mu$ in $\mathcal{P}_1(\mathbb{R}^d)$ if one of the equivalent statements of Lemma 4.8 is satisfied.

Proof of Lemma 4.8. „(4)⇒(1)“: Let $\varphi \in C_b^0(\mathbb{R}^d)$. Then φ satisfies $|\varphi(x)| \leq C(1+|x|)$ for $C := \sup_{x\in\mathbb{R}^d} |\varphi(x)|$ thus the claim of (1) follows.

„(1) ⇒ (2)“: This implication is obvious because if the limit exists, limes superior and the limes coincide thus giving the desired inequality.

„(2)⇒ (3)": Let $R > 0$. Then it holds that

$$\begin{aligned}\limsup_{k\to+\infty}\int_{|x|\geq R}|x|\,\mathrm{d}\mu_k &\leq \limsup_{k\to+\infty}\int_{\mathbb{R}^d}|x|\,\mathrm{d}\mu_k(x) - \liminf_{k\to+\infty}\int_{|x|\leq R}|x|\,\mathrm{d}\mu_k(x)\\ &\overset{(2)}{\leq} \int_{\mathbb{R}^d}|x|\,\mathrm{d}\mu_k(x) - \liminf_{k\to+\infty}\int_{|x|\leq R}|x|\,\mathrm{d}\mu_k(x)\,.\end{aligned}$$

We now fix $\epsilon > 0$ arbitrary and consider a function $f_{\epsilon,R} \in C_b^0(\mathbb{R}^d)$ such that

$$\mathbb{1}_{B_{R-\epsilon}(0)}(x) \leq f_{\epsilon,R}(x) \leq \mathbb{1}_{B_R(0)}(x) \quad \text{for all } x \in \mathbb{R}^d. \tag{4.2}$$

By our definition of $f_{\epsilon,R}$ we have that $x \mapsto f_{\epsilon,R}(x)\cdot|x|$ is a nonnegative, bounded and continuous function. This together with $\mu_k \rightharpoonup \mu$ as $k \to +\infty$ implies

$$\begin{aligned}\liminf_{k\to+\infty}\int_{|x|\leq R}|x|\,\mathrm{d}\mu_k(x) &\geq \liminf_{k\to+\infty}\int_{\mathbb{R}^d} f_{\epsilon,R}(x)\cdot|x|\,\mathrm{d}\mu_k(x) = \int_{\mathbb{R}^d} f_{\epsilon,R}(x)\cdot|x|\,\mathrm{d}\mu(x)\\ &\geq \int_{|x|\geq R-\epsilon}|x|\,\mathrm{d}\mu(x) \xrightarrow[R\to+\infty]{\substack{\text{Dominated}\\ \text{Convergence}}} \int_{\mathbb{R}^d}|x|\,\mathrm{d}\mu(x)\,.\end{aligned}$$

We thus get with the first estimate that

$$\limsup_{R\to+\infty}\limsup_{k\to+\infty}\int_{|x|\geq R}|x|\,\mathrm{d}\mu_k(x) \leq \int_{\mathbb{R}^d}|x|\,\mathrm{d}\mu(x) + \limsup_{R\to+\infty}\left(-\int_{|x|\leq R-\epsilon}|x|\,\mathrm{d}\mu(x)\right) = 0$$

„(3) ⇒ (4)": Choose a function $\varphi \in C^0$ satisfying (4) and define $f_{\epsilon,R}$ as in (4.2) for $R > 0$. Using

$$|1 - f_{\epsilon,R}(x)| = 1 - f_{\epsilon,R}(x) \leq 1 - \mathbb{1}_{B_{R-\epsilon}(0)}(x) \quad \text{for any } x \in \mathbb{R}^d$$

we obtain thanks to the weak convergence and the fact that $f_{\epsilon,R}(x)\cdot|x| \in C_b^0(\mathbb{R}^d)$ that

$$\begin{aligned}\left|\int_{\mathbb{R}^d}\varphi\,\mathrm{d}\mu - \int_{\mathbb{R}^d}\varphi\,\mathrm{d}\mu_k\right| \leq \underbrace{\left|\int_{\mathbb{R}^d}\varphi f_{\epsilon,R}\,\mathrm{d}\mu - \int_{\mathbb{R}^d}\varphi f_{\epsilon,R}\,\mathrm{d}\mu_k\right|}_{\xrightarrow{k\to+\infty}0} &+ \int_{\mathbb{R}^d}|x|\cdot\underbrace{|1 - f_{\epsilon,R}(x)|}_{\leq 1-\mathbb{1}_{B_{R-\epsilon}(0)}}\,\mathrm{d}\mu(x)\\ &+ \int_{\mathbb{R}^d}|x|\cdot|1 - f_{\epsilon,R}(x)|\,\mathrm{d}\mu_k(x)\end{aligned}$$

Because the first term goes to zero it holds that

$$\begin{aligned}&\limsup_{k\to+\infty}\left|\int_{\mathbb{R}^d}\varphi\,\mathrm{d}\mu - \int_{\mathbb{R}^d}\varphi\,\mathrm{d}\mu_k\right|\\ &\qquad\leq C\int_{|x|\geq R-\epsilon}(1+|x|)\,\mathrm{d}\mu(x) + \limsup_{k\to+\infty} C\int_{|x|\geq R-\epsilon}(1-|x|)\,\mathrm{d}\mu_k(x)\end{aligned}$$

As the LHS is independent of R we can just take the limit $R \to +\infty$ which causes the RHS to converge to zero from which (4) follows. This concludes the proof. □

Corollary 4.10. Let $(\mu_k)_{k\in\mathbb{N}}$ be a sequence of probability measures in $\mathcal{P}_1(\mathbb{R}^d)$ and $\mu \in \mathcal{P}_1(\mathbb{R}^d)$. The following statements are equivalent:

(1) $\lim\limits_{k\to+\infty} \mathcal{W}_1(\mu_k, \mu) = 0$.

(2) $\mu_k \rightharpoonup \mu$ and $\lim_{k\to+\infty} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x) = \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x)$

Our goal will be to show that $(\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$ is a complete metric space. To this end we want to introduce a criterion which gives the weak convergence of sequence of probability measures called *tightness*. But first let us discuss what can go wrong:

Example 4.11. Consider the sequence $(\delta_n)_{n\in\mathbb{N}}$ where δ_p denotes the Dirac delta of a point $p \in \mathbb{R}^d$ (in this case $d = 1$). For a bounded and continuous function f we obtain

$$\int_{\mathbb{R}} f \,\mathrm{d}\delta_n = f(n)$$

thus δ_n cannot converge weakly. $\triangle$

In order to prevent the mass from escaping to $+\infty$ one can introduce the notion of tightness.

Definition 4.12. A family of probability measure $\mathcal{F} \subseteq \mathcal{P}(\mathbb{R}^d)$ is **tight** if and only if for any $\epsilon > 0$ we find $K \subset \mathbb{R}^d$ compact such that $\mu(K) > 1 - \epsilon$ for any $\mu \in \mathcal{F}$.

This definition ensures that all the mass is concentrated on compact sets. Also for this reason the previous example does not fit this definition. The notion of tightness ensures the existens of weakly convergent subsequences.

Theorem 4.13 (Prokhorov)**.** Let $(\mu_n)_{n\in\mathbb{N}}$ be a tight sequence of probability measures on $\mathbb{R}^d$. Then there exists a weakly convergent subsequence of this sequence i.e., we find $\mu \in \mathcal{P}(\mathbb{R}^d)$ and $(\mu_{n_\ell})_{\ell\in\mathbb{N}}$ with $\mu_{n_\ell} \overset{\ell\to+\infty}{\rightharpoonup} \mu$.

This allows us achieve our goal:

Theorem 4.14. $(\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$ is a complete metric space.

Proof. We already proved that $\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$ is indeed a metric space; it is thus left to show the completeness of said space. Let $(\mu_k)_{k\in\mathbb{N}}$ be a Cauchy sequence in $\mathcal{P}_1(\mathbb{R}^d), \mathcal{W}_1)$. We shall prove that this sequence converges.

Step 1: $(\mu_k)_{k\in\mathbb{N}}$ is tight. Let $N \in \mathbb{N}$ such that for any $k, \ell > N$ it holds that $\mathcal{W}_1(\mu_k, \mu_\ell) < 1$. Since the Euclidean norm on $\mathbb{R}^d$ is a Lipschitz continuous function we get for $k > N$ that

$$\begin{aligned}\int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x) &= \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x) \pm \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{N+1}(x)\\ &\le \mathcal{W}_1(\mu_k, \mu_{N+1}) + \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{N+1}(x)\\ &\le 1 + \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{N+1}(x) < +\infty,\end{aligned}$$

thus if we take the supremum over all $k \in \mathbb{N}$ we see that

$$\sup_{k\in\mathbb{N}} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x) \le \max\left\{\max_{k=1,\dots,N} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_k(x)\,, 1 + \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{N+1}(x)\right\} =: C < +\infty.$$

For $R > 0$, we get the following lower bound:

$$1 \geq \inf_{k\in\mathbb{N}} \mu(B_R) = \inf_{k\in\mathbb{N}} \int_{|x|\leq R} \mathrm{d}\mu_k(x) = 1 - \sup_{k\in\mathbb{N}} \int_{|x|\geq R} \mathrm{d}\mu_k(x)$$
$$\geq 1 - \sup_{k\in\mathbb{N}} \int_{\mathbb{R}^d} \frac{|x|}{R} \,\mathrm{d}\mu_k(x) \geq 1 - \frac{C}{R}$$

and thus deduce that the sequence $(\mu_k)_{k\in\mathbb{N}}$ is tight as $B_R(0)$ is a compact set for any $R > 0$. By choosing R big, we can get the bound arbitrarily close to 1.

Step 2: There exists a weakly convergent subsequence. By Theorem 4.13 we find a weakly convergent subsequence $(\mu_{k_\ell})_{\ell\in\mathbb{N}}$ and a limit $\mu \in \mathcal{P}(\mathbb{R}^d)$ such that $\mu_{k_\ell} \rightharpoonup \mu$ in $\mathcal{P}(\mathbb{R}^d)$ as $\ell \to +\infty$. For a suitable $\epsilon > 0$ take $f_{\epsilon,R}$ as in (4.2). We then obtain since $x \mapsto f_\epsilon(x)|x|$ is a bounded and continuous function that

$$\int_{|x|\leq R-\epsilon} |x| \,\mathrm{d}\mu(x) \leq \int_{\mathbb{R}^d} |x| f_\epsilon(x) \,\mathrm{d}\mu(x) = \lim_{\ell\to+\infty} \int_{\mathbb{R}^d} |x| f_\epsilon(x) \,\mathrm{d}\mu_{k_\ell}(x)$$
$$\leq \limsup_{\ell\to+\infty} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{k_\ell}(x) \leq C$$

which implies that

$$\int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x) \overset{\substack{\text{Monotone}\\ \text{Convergence}}}{=} \lim_{R\to+\infty} \int_{|x|\leq R-\epsilon} |x| \,\mathrm{d}\mu(x) \leq C < +\infty,$$

i.e., $\mu \in \mathcal{P}_1(\mathbb{R}^d)$ as it has finite first moment.

Step 4: $\mathcal{W}_1(\mu_k, \mu) \xrightarrow{k\to+\infty} 0$. For a fixed R we define a function $g_R : \mathbb{R}^d \to \mathbb{R}$ by setting

$$g_R(x) := \begin{cases} |x|, & |x| \leq R \\ 2R - |x|, & R < |x| \leq 2R \\ 0, & \text{otherwise} \end{cases}.$$

One can easily check that g_N is Lipschitz continuous and satisfies

$$|x|\mathbb{1}_{B_R(0)}(x) \leq g_R(x) \leq |x| \quad \text{for any } x \in \mathbb{R}^d \text{ and} \quad \|g_R\|_{\mathrm{Lip}(\mathbb{R}^d)} \leq 1.$$

For an arbitrary $\delta > 0$ we choose $n, \ell \in \mathbb{N}$ such that $\mathcal{W}_1(\mu_{k_\ell}, \mu_{k_n}) < \delta$ and obtain

$$\begin{aligned}\int_{|x|\leq R} |x| \,\mathrm{d}\mu_{k_\ell}(x) &\leq \int_{\mathbb{R}^d} g_R \,\mathrm{d}\mu_{k_\ell} \\ &\leq \mathcal{W}_1(\mu_{k_\ell}, \mu_{k_n}) + \int_{\mathbb{R}^d} g_R \,\mathrm{d}\mu_{k_n} \\ &\leq \delta + \underbrace{\int_{\mathbb{R}^d} g_R \,\mathrm{d}\mu_{k_n} - \int_{\mathbb{R}^d} g_R \,\mathrm{d}\mu}_{\xrightarrow{n\to+\infty} 0 \text{ (weak convergence)}} + \int_{\mathbb{R}^d} g_R \,\mathrm{d}\mu\end{aligned}$$

By letting $R, n \to +\infty$ in the above expression we find

$$\int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{k_\ell} = \lim_{R\to+\infty} \int_{|x|\leq R} |x| \,\mathrm{d}\mu_{k_\ell}(x) \leq \delta + \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x) \quad \text{for } \ell \text{ large enough.}$$

Using the arbitrariness of $\delta > 0$ we deduce that

$$\lim_{\ell\to+\infty} \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu_{k_\ell} \leq \int_{\mathbb{R}^d} |x| \,\mathrm{d}\mu(x)\,.$$

By applying Lemma 4.8 we see that $(\mu_{k_\ell})_{\ell\in\mathbb{N}}$ converges weakly to μ in $\mathcal{P}_1(\mathbb{R}^d)$. Using Corollary 4.10 we deduce that $\mathcal{W}_1(\mu_{k_\ell}, \mu) \to 0$ as $\ell \to +\infty$. By the uniqueness of the limit the whole sequence goes to zero (We leave this fact as an exercise for the reader to prove). □

5. The Vlasov Equation: Solutions

5.1. Introduction and Notation

In the following we study the so-called *Vlasov equation*: We seek a differentiable function $f : \mathbb{R}_+ \times \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$, $(t, x, v) \mapsto f(t, x, v)$ solving

$$\begin{cases} \partial_t f + v \cdot \nabla_x f + E \cdot \nabla_v f = 0 \\ E(t,x) = \Big(\nabla_x V \star \rho(t,\cdot)\Big)(x) \\ \rho(t,x) = \int_{\mathbb{R}^d} f(t,x,v)\,\mathrm{d}v \\ f(0,x,v) = f_0(x,v) \end{cases} . \tag{5.1}$$

where $V : \mathbb{R}^d \to \mathbb{R}$, $f_0 : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ are given functions. In plasma physics we call $V(x) = \frac{1}{|x|}$ the *Coulomb-potential* and in Astro-physics we call $V(x) = -\frac{1}{|x|}$ the *gravitational potential*. In both cases, $V = \pm\frac{1}{|x|}$ is the *Green's* function of the Laplacian operator $\Delta = \sum_i \frac{\partial}{\partial x_i}$ i.e.,

$$\begin{cases} E(t,x) = -\nabla_x \phi(t,x) \\ \pm 4\pi \Delta_x \phi(t,x) = \rho(t,x) \end{cases} .$$

This above equation is called the *Poisson-equation*. It is the reason we call (5.1) the *Vlasov-Poisson-equation*. When V is the Coulomb-potential, the function E models electrostatic interaction between particles which explains the notation E (we call E the electric field). Contrary to the transport equation we now have a force field depending on unknowns and thus are dealing with a nonlinear equation. For simplicity we will from now on consider $d = 3$ and introduce the following notations:

Notation 5.1. We write $K(x) := -\nabla V(x)$ and consider a family of measures $\mu = (\mu_t)_{t\in\mathbb{R}_+}$ given by the density f i.e.,

$$\mu_t(B) := \int_B f(t,x,v)\,\mathrm{d}x\,\mathrm{d}v \quad \text{for any } B \in \mathcal{B}(\mathbb{R}^3 \times \mathbb{R}^3)$$

We then can write the function E in the following way:

$$\begin{aligned} E(t,x) &= -\Big((-\nabla V) \star \rho(t,\cdot)\Big)(x) = -\int_{\mathbb{R}^3} K(x-y)\rho(t,y)\,\mathrm{d}y \\ &= -\int_{\mathbb{R}^3} K(x-y) \int_{\mathbb{R}^3} f(t,y,v)\,\mathrm{d}v\,\mathrm{d}y = -\int_{\mathbb{R}^6} K(x-y)\,\mathrm{d}\mu_t(y,v) \end{aligned} \tag{5.2}$$

This suggests to introduce the notation

$$E_{\mu_t}(x) := -\int_{\mathbb{R}^6} K(x-y)\,\mathrm{d}\mu_t(y,v)\,.$$

Using the above notation the Vlasov equation becomes

$$\begin{cases} \partial_t \mu_t + V\nabla_x \mu_t + E_{\mu_t} \cdot \nabla_v \mu_t = 0 \\ \mu_t\Big|_{t=0} = \mu_0 \end{cases} \qquad \text{where } \mu_0 \text{ is a given Borel measure.}$$

The goal will be to show existence and uniqueness of measure valued solutions and classical solutions. We will use the notion of measure valued solutions from the famous work of *Dobrushin*.

Definition 5.2. A family of measures on $\mathbb{R}^3 \times \mathbb{R}^3$ is a solution to 5.1 if and only if for any $h \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R}^3)$ the following two conditions are satisfied:

(1) The map

$$t \mapsto \int_{\mathbb{R}^3 \times \mathbb{R}^3} h(x,v)\,\mathrm{d}\mu_t(x,v) \in C^1([0,T];\mathbb{R})$$

is continuously differentiable and

(2) It holds that

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^3 \times \mathbb{R}^3} h(x,v)\,\mathrm{d}\mu_t(x,v) = \int_{\mathbb{R}^3 \times \mathbb{R}^3} v \cdot \nabla_x h(x,v) + E_{\mu_t}(x) \cdot \nabla_v h(x,v)\,\mathrm{d}\mu_t(x,v)\,.$$

Remark 5.3. One can also consider signed measures, for the sake of simplicity we will not do this. △

Notation 5.4. For a measure μ and an integrable function h we will write

$$\mu(h) := \int_{\mathbb{R}^6} h(x,z)\,\mathrm{d}\mu(x,z)$$

to simplify the notation. If we group x and v together in one variable gets us

$$\underbrace{G^\mu(t,z)}_{\in \mathbb{R}^6} \overset{z:=(x,v)}{:=} \begin{bmatrix} v \\ E_{\mu_t}(x) \end{bmatrix},$$

thus the transport equation then reads as

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t} \mu_t(h) = \mu_t(G^\mu(t,\cdot) \cdot \nabla_z h) \\ \mu_t\Big|_{t=0} = \mu_0 \end{cases} \qquad \text{where } h = h(z) \in C_c^\infty(\mathbb{R}^6). \tag{5.3}$$

5.2. Existence and Uniqueness of measure valued solutions to the Vlasov-Equation

We achieve our goal of proving the existence and uniqueness of measure valued solutions to the Vlasov-equation in two steps. Namely,

(1) Show the existence and uniqueness of a solution to the corresponding linear equation and

(2) Show that the Solution to the Vlasov-equation is the fixed point of an operator defined by the linear equation and use the Contraction mapping principle to conclude.

5.2.1. Solutions to the linear equation

For now, we are interested in the Cauchy problem

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t}\mu_t(h) = \mu_t(G(t,\cdot)\cdot\nabla_z h) \\ \mu_t(h)\Big|_{t=0} = \mu_0(h) \end{cases} \quad \text{for any } h \in C_c^\infty(\mathbb{R}^6) \tag{5.4}$$

with $G = G(t,z)$ a given time dependent vector field on $\mathbb{R}^6$. Contrary to (5.3) the vector field does not depend on the solution μ_t making this problem much easier to deal with. If the vector field G is of the form

$$G(z,t) = \begin{bmatrix} v \\ a(t,x,v) \end{bmatrix}, \quad z = (x,v), \quad x, v \in \mathbb{R}^3$$

we call the equation (5.4) *Liouville-equation*.

Remark 5.5. Actually Theorems 2.1 and 2.3 hold under more general less restrictive assumptions on the vector field b (see Golse, Thm 2.2.2 and Thm 2.2.3), namely we require

(H1) $b \in C^0$ and $\nabla_x b \in C([0,T]\times\mathbb{R}^d;\mathbb{R}^{d\times d})$ where $\nabla_x b$ denotes the Jacobian matrix in space and

(H2) the existence of a constant $\tilde{C} > 0$ such that $|b(t,x)| \le C(1+|x|)$ for any x and t.

△

Lemma 5.6. Let $T > 0$, $G : [0,T]\times\mathbb{R}^6 \to \mathbb{R}^6$ satisfying conditions (H1) and (H2) and $\operatorname{div} G(t,z) = 0$ for any $t \in [0,T]$ and μ_0 a finite Borel-measure on $\mathbb{R}^6$. Then (5.4) has a unique solution satisfying

$$\sup_{t\in[0,T]} \sup_{B\in\mathcal{B}(\mathbb{R}^6)} \mu_t(B) \le C_T < +\infty$$

where C_T is a constant depending on T. Moreover the solution is given by the push forward of the initial datum μ_0 by the flow Z associated to the vector field G i.e.,

$$\mu_t = Z(t,0,\cdot)_{\#}\mu_0$$

where Z is the flow of G.

Proof. We start by first proving the *existence* of a solution and consider the characteristic flow of the vector field G satisfying the equation

$$\begin{cases} \dfrac{\partial}{\partial t} Z(t,s,z) = G(t, Z(t,s,z)) \\ Z(s,s,z) = z \end{cases}.$$

Note that the flow $Z \in C1([0,T]\times[0,T]\times\mathbb{R}^6;\mathbb{R}^6)$ is uniquely determined by this equation and its existence is guarantied which is ensured by Theorems 2.1 and 2.3 and the previous remark. Similarly as in the proof of Theorem 3.7 we compute for a test function $h \in C_c^\infty(\mathbb{R}^6)$,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} Z(t,0,\cdot)_\# \mu_0(h) &= \frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^6} h(z)\,\mathrm{d}(Z(t,0,\cdot)_\# \mu_0)(z) \\ &= \frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^6} h(Z(t,0,z))\,\mathrm{d}\mu_0(z) \\ &= \int_{\mathbb{R}^6} (\nabla_z h) \circ Z(t,0,z) \cdot G(t, Z(t,0,z))\,\mathrm{d}\mu_0(z) \\ &= Z(t,0,\cdot)_\# \mu_0(G(t,\cdot) \cdot \nabla_z h).\end{aligned}$$

Since μ_0 is finite and a Borel measure we obtain that

$$\sup_{t \in [0,T]} \sup_{B \in \mathcal{B}(\mathbb{R}^6)} Z(t,0,\cdot)_\# \mu_0(B) < +\infty$$

which implies that $\mu_t = Z(t,0,\cdot)_\# \mu_0$ is indeed a solution.

To prove *uniqueness* assume that $(\mu_t)_{t\in[0,T]}$ is a solution in the sense of Lemma 5.6. Choose $h \in C_c^1(\mathbb{R}^6)$ and a sequence $(h_n)_{n\in\mathbb{N}} \in (C_c^\infty(\mathbb{R}^6))^{\mathbb{N}}$ such that

$$\sup_{x\in\mathbb{R}^6} \Big(|h_n(x) - h(x)| + |\nabla_z h_n(x) - \nabla_z h(x)| \Big) \xrightarrow{n\to+\infty} 0.$$

For any $n \in \mathbb{N}$ it holds that

$$\frac{\mathrm{d}}{\mathrm{d}t} \mu_t(h_n) = \mu_t(G(t,\cdot) \cdot \nabla_z h).$$

This implies that for any $n \in \mathbb{N}$,

$$\mu_t(h_n) = \mu_0(h_n) + \int_0^t \mu_s(G(s,\cdot) \cdot \nabla_z h_n)\,\mathrm{d}s\,.$$

Using that h and h_n have uniform compact support we obtain by letting $n \to +\infty$

$$\mu_t(h) = \mu_0(h) + \int_0^t \mu_s(G(s,\cdot) \cdot \nabla_z h)\,\mathrm{d}s\,.$$

Let $K \subseteq \mathbb{R}^6$ be a compact set such that $\operatorname{spt} h_n \subseteq K$ and $h \subseteq K$. We then get the following estimate:

$$\begin{aligned}\Big|\mu_t(G(t,\cdot)\cdot\nabla_z h) - \mu_s(G(t,\cdot)\cdot\nabla_z h)\Big| &\le \Big|\mu_t(G(t,\cdot)\cdot\nabla_z h_n) - \mu_s(G(t,\cdot)\cdot\nabla_z h_n)\Big| \\ &\quad + \underbrace{\sup_{t\in[0,T]} |\mu_t(G(t,\cdot)\cdot\nabla_z h) - \mu_s(G(t,\cdot)\cdot\nabla_z h_n)|}_{=2\sup_{t\in[0,T]} \int_{\mathbb{R}^6} G(t,z)\cdot\nabla_z(h-h_n)(z)\mathrm{d}\mu_t(z)} \\ &\le \sup_{t\in[0,T]} \underbrace{\|\nabla_z(h-h_n)\|_{L^\infty(K)}}_{\xrightarrow{n\to+\infty}0} \|G(t,\cdot)\|_{L^\infty(K)} \underbrace{\int_K \mathrm{d}\mu_t}_{<+\infty}\,.\end{aligned}$$

We therefore see that $t \mapsto \mu_t(G(t,\cdot)\cdot\nabla_z h)$ is a continuous function hence we can differentiate the integral. This implies that $t \mapsto \mu_t(h)$ is continuously differentiable and satisfies our linear equation. We now define $h_t(z) := h(Z(0,t,z))$. From the previous results we know that h_t is infinitely often differentiable and satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t} h_t(z) = -G(t,z) \cdot \nabla_z h_t(z).$$

We thus see that

$$\frac{\mathrm{d}}{\mathrm{d}t}h_t(z) \overset{\substack{\text{Product}\\\text{Rule}}}{=} \dot{\mu}_t(h_t) + \mu_t(\dot{h}_t) = \mu_t(G(t,\cdot)\cdot\nabla_z h_t) - \mu_t(G(t,\cdot)\cdot\nabla_z h_t) = 0$$

and therefore for any $h \in C_c^\infty(\mathbb{R}^6)$,

$$\int_{\mathbb{R}^6} h(z)\,\mathrm{d}\mu_0(z) = \int_{\mathbb{R}^6} h(Z(0,t,z))\,\mathrm{d}\mu_t(z)$$

thus the uniqueness follows. □

5.2.2. Solutions of the general Equation

The goal is to prove the existence and uniqueness of solutions to the Vlasov equation (5.1). We will do this by a fixed point argument i.e., by constructing a contraction where every fixed point solves the (5.1) and use the contraction mapping principal to show that a unique fixed point exists.

In order to do that we recall the equation

$$\begin{cases} \frac{\mathrm{d}}{\mathrm{d}t}\mu_t(h) = \mu_t(G^\nu(t,\cdot)\cdot\nabla_z h_t(z)) & t\in[0,T] \\ \mu_t(h) = \mu_0(h) \end{cases}, \quad h \in C_c^\infty(\mathbb{R}^6) \tag{5.5}$$

where for $\nu = (\nu_t)_{t\in[0,T]}$ a family of measures in $\mathcal{P}_1(\mathbb{R}^d)$

$$G^\nu(t,z) = \begin{bmatrix} v \\ E_{\nu_t}(x) \end{bmatrix} \qquad z=(x,v)\in\mathbb{R}^3\times\mathbb{R}^3.$$

For a given $\mu_0 \in \mathcal{P}_1(\mathbb{R}^6)$ we define

$$\mathfrak{m}_{T,\mu_0} := \left\{ (\nu_t)_{t\in[0,T]} \subseteq (\mathcal{P}_1(\mathbb{R}^6)) \,\middle|\, \begin{matrix} \nu_t|_{t=0} = \mu_0, K\in C_b^1(\mathbb{R}^3) \\ \forall x\in\mathbb{R}^6, [t\in[0,T]\mapsto E_{\nu_t}(x)\in\mathbb{R}]\in C^0 \end{matrix} \right\}$$

where $K:\mathbb{R}^3\to\mathbb{R}^3$ is defined as in (5.2) and fixed.

Lemma 5.7. Take $K\in C_b^0(\mathbb{R}^3)$, where K is given by (5.2). Let $\nu = (\nu_t)_{t\in[0,T]} \in \mathfrak{m}_{T,\mu_0}$ where μ_0 is given and G^ν defined in the usual way:

$$G^\nu(t,z) = \begin{bmatrix} v \\ E_{\nu_t}(x) \end{bmatrix}, \quad z=(x,v)\in\mathbb{R}^3\times\mathbb{R}^3.$$

Then $\operatorname{div}_z G^\nu(t,z) = 0$ for any (t,z) and the conditions (H1) and (H2) are satisfied.

Proof. By differentiating V with respect to x and G_{ν_t} with respect to v we obtain that $\operatorname{div}_z G^\nu(t,z)=0$. As K is bounded we get that

$$|E_{\nu_t}(x)| = \left|\int_{\mathbb{R}^3} K(x-y)\,\mathrm{d}\nu_t(y,z)\right| \le \|K\|_{L^\infty(\mathbb{R}^3)} \underbrace{\nu_t(\mathbb{R}^3)}_{=1}$$

and thus with the usual notation $z=(x,v)$ we can estimate component wise with

$$|G^\nu(t,z)| \le \underbrace{|v|}_{\le|z|} + |E_{\nu_t}(x)| \le C(1+|z|) \quad \text{for any } t\in[0,T]$$

which implies (H2).

Let $\epsilon > 0$ be an arbitrary number. As K has a continuous and bounded derivative we find $\delta_\epsilon > 0$ with

$$|K(x-y) - K(x'-y)| < \epsilon \quad \text{for any } x, x' \in \mathbb{R}^3 \text{ with } |x - x'| < \delta_\epsilon$$

thus for $|x - x'| < \delta_\epsilon$ one sees that

$$\sup_{t\in[0,T]} |E_{\nu_t}(x) - E_{\nu_t}(x')| \leq \epsilon \cdot \sup_{t\in[0,T]} \nu_t(\mathbb{R}^6) \leq \epsilon$$

which implies that

$$\lim_{x'\to x} \sup_{t\in[0,T]} |E_{\nu_t}(x) - E_{\nu_t}(x')| = 0 \quad \text{for any } x \in \mathbb{R}^3.$$

Using that E_{ν_t} is continuous in t we obtain that G^ν is a continuous function. This is a consequence of

$$\begin{aligned} |G^\nu(t,z) - G^\nu(t',z')| &\leq |v - v'| + |E_{\nu_t}(x) - E_{\nu_{t'}}(x')| \\ &\leq |v - v'| + |E_{\nu_t}(x) - E_{\nu_{t'}(x)}| + |E_{\nu_{t'}}(x) - E_{\nu_{t'}}(x')| \xrightarrow{(t',x')\to(t,x)} 0. \end{aligned}$$

Similarly it follows that

$$\lim_{x'\to x} \sup_{t\in[0,T]} |\nabla_x E_{\nu_t}(x) - \nabla_x E_{\nu_t}(x')|.$$

Moreover note that for any $\epsilon > 0$ there exists $\delta_\epsilon > 0$ such that

$$\left| \frac{K(x+\Delta x) - K(x)}{\Delta x} - \nabla_x K(x) \right| < \epsilon \quad \text{for any } |\Delta x| < \delta_\epsilon.$$

which implies that

$$\lim_{\Delta x\to 0} \sup_{t\in[0,T]} \left| \frac{E_{\nu_t}(x+\Delta x) - E_{\nu_t}(x)}{\Delta x} - \nabla_x E_{\nu_t}(x) \right| = 0$$

and thus $t \mapsto \nabla_x E_{\nu_t}(x)$ is a continuous function for any $x \in \mathbb{R}^3$ because

$$\begin{aligned} |\nabla_x E_{\nu_t(x)} - \nabla_x E_{\nu_s}(x)| \leq 2 \cdot \sup_{t\in[0,T]} &\left| \frac{E_{\nu_t}(x+\Delta x) - E_{\nu_t}(x)}{\Delta x} - \nabla_x E_{\nu_t}(x) \right| \\ &+ \frac{|E_{\nu_t}(x+\Delta x) + E_{\nu_s}(x+\Delta x)| + |E_{\nu_t}(x) - E_{\nu_s}(x)|}{\Delta x} \end{aligned}$$

The continuity follows from chosing $|\Delta x|$ small and letting $t \to s$ which works because the electric field $E_{\nu_t}(x)$ is continuous with respect to time variable t. Thanks to the above computations we deduce that (H1) must hold as well. This concludes the proof. □

Next we define the following mapping:

$$\Phi : \mathfrak{m}_{T,\mu_0} \to \mathfrak{m}_{T,\mu_0}, \quad (\nu_t)_{t\in[0,T]} \mapsto (\mu_t)_{t\in[0,T]}$$

where $(\mu_t)_{t\in[0,T]}$ is the measure valued solution to the Vlasov equation (5.5). Moreover we need a metric on the space $\mathfrak{m}_{T,\mu_0}$ so we define

$$\tilde{\mathcal{W}}_1\Big((\mu_t)_{t\in[0,T]}, (\nu_t)_{t\in[0,T]}\Big) := \sup_{t\in[0,T]} \mathcal{W}_1(\mu_t, \nu_t).$$

We will later show that Φ is a contraction on the metric space $(\mathfrak{m}_{T,\mu_0}, \tilde{\mathcal{W}}_1)$ which is going to allow us to use Banach's fixed point argument. But first we need to show the following:

Lemma 5.8. (1) Φ is well defined.

(2) $(\mathfrak{m}_{T,\mu_0}, \tilde{\mathcal{W}}_1)$ is indeed a complete metric space.

Proof. (1) We leave the proof of the fact that Φ is well defined as an exercise for the reader.

(2) To prove completeness we take $((\nu_t^n)_{t\in[0,T]})_{n\in\mathbb{N}}$ a Cauchy sequence in $(\mathfrak{m}_{T,\mu_0}, \tilde{\mathcal{W}}_1)$. We note that for any $t \in [0,T]$ the sequence $(\nu_t^n)_{n\in\mathbb{N}}$ is a Cauchy sequence and converges by the completeness of $(\mathcal{P}_1(\mathbb{R}^6), \mathcal{W}_1)$ which we proved in theorem 4.14. We thus deduce that there exists a limit measure ν_t for any $t \in [0,T]$ with

$$\tilde{\mathcal{W}}_1\Big((\nu_t^n)_{t\in[0,T]}, (\nu_t)_{t\in[0,T]}\Big) = \sup_{t\in[0,T]} \mathcal{W}_1(\nu_t^n, \nu_t) \xrightarrow{n\to+\infty} 0$$

and $\nu_0^n = \mu_0$ for any $n \in \mathbb{N}$ so $\nu_t|_{t=0} = \mu_0$. It is therefore left to prove that $(\nu_t)_{t\in[0,T]}$ is indeed an element of $\mathfrak{m}_{T,\mu_0}$. Using that $K \in C_b^1(\mathbb{R}^3)$ we have that K is Lipschitz:

$$\begin{aligned}|K(x) - K(y)| &= \left|\int_0^1 \frac{\mathrm{d}}{\mathrm{d}\lambda} K(\lambda y + (1-\lambda)x)\,\mathrm{d}\lambda\right| \\ &= \left|\int_0^1 (\nabla K)(\lambda y + (1-\lambda)x)\cdot(y-x)\,\mathrm{d}\lambda\right| \le \|\nabla K\|_{L^\infty(\mathbb{R}^3)}|x-y|.\end{aligned}$$

This implies that the function $(x,y) \mapsto K(x-y)/\|\nabla K\|_{L^\infty(\mathbb{R}^3)}$ is bounded and Lipschitz continuous with Lipschitz constant smaller or equal than one, so

$$\begin{aligned}\sup_{t\in[0,T]}\Big|E_{\nu_t^n}(x) - E_{\nu_t}(x)\Big| &\le \sup_{t\in[0,T]}\left|\int_{\mathbb{R}^3} \frac{K(x-y)}{\|\nabla K\|_{L^\infty(\mathbb{R}^3)}}\cdot\|\nabla K\|_{L^\infty(\mathbb{R}^3)}\Big(\mathrm{d}\nu_t^n(y,x) - \mathrm{d}\nu_t(y,x)\Big)\right| \\ &\le \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \sup_{t\in[0,T]} \mathcal{W}_1(\nu_t^n, \nu_t).\end{aligned}$$

Consequently,

$$|E_{\nu_t}(x) - E_{\nu_s}(x)| \le \underbrace{|E_{\nu_t^n}(x) - E_{\nu_s^n}(x)|}_{\xrightarrow{|t-s|\to 0} 0} + \underbrace{2\cdot\sup_{t\in[0,T]}|E_{\nu_t^n}(x) - E_{\nu_t}(x)|}_{\xrightarrow{n\to+\infty} 0}$$

from which we deduce that the map $t \mapsto E_{\nu_t}(x)$ is continuous for any $x \in \mathbb{R}^6$. This concludes the proof. $\square$

Remark 5.9. The estimate

$$|E_{\nu_t}(x) - E_{\nu_{t'}}(x)| \le \|\nabla K\|_{L^\infty(\mathbb{R}^6)}\mathcal{W}_1(\nu_t, \nu_{t'})$$

shows that the Wasserstein distance is suitable to controll the forces of different solutions. This fact will be one of the key elements in Dobrushin's proof. $\triangle$

Lemma 5.10. For any $K \in C_b^1(\mathbb{R}^3)$ there exists $\delta > 0$ such that for any $T < \delta$, Φ is a contraction on $(\mathfrak{m}_{T,\mu_0}, \tilde{\mathcal{W}}_1)$.

Proof. Let Z^ν be the flow of G^ν i.e.,

$$\begin{cases} \dfrac{\partial}{\partial s} Z^\nu(s,t,x) = G^\nu(s, Z^\nu(s,t,x)) \\ Z^\nu(t,t,x) = x \end{cases} \quad \text{for any } s,t \in [0,T], x \in \mathbb{R}^6$$

and $Z^{\tilde{\nu}}$ be the flow of $G^{\tilde{\nu}}$. Then for any $t \in [0,T]$ we obtain

$$\begin{aligned} |Z^\nu(t,0,z) - Z^{\tilde{\nu}}(t,0,z)| &\le \int_0^t |G^\nu(s, Z^\nu(s,0,z)) - G^{\tilde{\nu}}(s, Z^{\tilde{\nu}}(s,0,z))| \,\mathrm{d}s \\ &\le \int_0^t |G^\nu(s, Z^\nu(s,0,z)) - G^\nu(s, Z^{\tilde{\nu}}(s,0,z))| \,\mathrm{d}s \\ &\quad + \int_0^t |G^\nu(s, Z^{\tilde{\nu}}(s,0,z)) - G^{\tilde{\nu}}(s, Z^{\tilde{\nu}}(s,0,z))| \,\mathrm{d}s \\ &=: \mathrm{I}_t + \mathrm{II}_t. \end{aligned}$$

We write the flow as $Z^\nu = \begin{bmatrix} V^\nu & X^\nu \end{bmatrix}^\top \in \mathbb{R}^3 \times \mathbb{R}^3$ (and analog for the flow $Z^{\tilde{\nu}}$) so

$$G^\nu(t, Z^{\tilde{\nu}}(t,0,z)) = \begin{bmatrix} V^{\tilde{\nu}}(t,0,z) \\ E_{\nu_t}(X^{\tilde{\nu}}(t,0,z)) \end{bmatrix}$$

Using that $|K(y) - K(x)| \le \|\nabla K\|_{L^\infty} |x-y|$ we see that

$$\begin{aligned} \mathrm{I}_t &\le \int_0^t |V^\nu(s,0,z) - V^{\tilde{\nu}}(s,0,z)| + |E_{\nu_s}(X^\nu(s,0,z)) - E_{\nu_s}(X^{\tilde{\nu}}(s,0,z))| \,\mathrm{d}s \\ &\le \int_0^t |Z^\nu(s,0,z) - Z^{\tilde{\nu}}(s,0,z)| + \|\nabla K\|_{L^\infty} ||X^\nu(s,0,z) - X^{\tilde{\nu}}(s,0,z)| \,\mathrm{d}s \\ &\le t(1 + \|\nabla K\|_{L^\infty(\mathbb{R}^3)}) \sup_{s \in [0,t]} |Z^\nu(s,0,z) - Z^{\tilde{\nu}}(s,0,z)|. \end{aligned}$$

For the second term we get

$$\begin{aligned} \mathrm{II}_t &\le \int_0^t |E_{\nu_s}(X^{\tilde{\nu}}(s,0,z)) - E_{\tilde{\nu}_s}(X^{\tilde{\nu}}(s,0,z))| \,\mathrm{d}s \\ &\le \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \sup_{s \in [0,T]} \mathcal{W}_1(\nu_s, \tilde{\nu}_s) = \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \tilde{\mathcal{W}}_1\Big((\nu_s)_{s \in [0,t]}, (\tilde{\nu}_s)_{s \in [0,t]} \Big). \end{aligned}$$

Moreover with $|E_{\nu_t}(x) - E_{\tilde{\nu}_t}(x)| \le \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \mathcal{W}_1(\nu_t, \tilde{\nu}_t)$ follows

$$\begin{aligned} |Z^\nu(t,0,z) - Z^{\tilde{\nu}}(t,0,z)| &\le t(1 + \|\nabla K\|_{L^\infty(\mathbb{R}^3)}) \sup_{s \in [0,T]} |Z^\nu(s,0,z) - Z^{\tilde{\nu}}(s,0,z)| \\ &\quad + t \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \underbrace{\sup_{s \in [0,T]} \mathcal{W}_1(\nu_s, \tilde{\nu}_s)}_{= \tilde{\mathcal{W}}_1((\nu_s)_{s \in [0,T]}, (\tilde{\nu}_s)_{s \in [0,T]})} \end{aligned}$$

and for

$$\alpha(\nu, \tilde{\nu}, z) := \sup_{s \in [0,T]} |Z^\nu(s,0,z) - Z^{\tilde{\nu}}(s,0,z)|$$

one thus sees

$$\alpha(\nu, \tilde{\nu}, z) \le T(1 + \|\nabla K\|_{L^\infty(\mathbb{R}^3)}) \alpha(\nu, \tilde{\nu}, z) + T \cdot \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \tilde{\mathcal{W}}_1\Big((\nu_t)_{t \in [0,T]}, (\tilde{\nu}_t)_{t \in [0,T]} \Big).$$

Taking $T < (1+\|\nabla K\|_{L^\infty(\mathbb{R}^3)})^{-1}$ we obtain that

$$\alpha(\nu,\tilde\nu,z) \le \frac{\|\nabla K\|_{L^\infty(\mathbb{R}^3)}}{1-(1+\|\nabla K\|_{L^\infty(\mathbb{R}^3)})\cdot T}\cdot \tilde{\mathcal{W}}_1((\nu_t)_{t\in[0,T]},(\tilde\nu_t)_{t\in[0,T]})$$

and

$$\begin{aligned}
\tilde{\mathcal{W}}_1\Big(\Phi((\nu_t)_t),\Phi((\tilde\nu_t)_t)\Big) &= \sup_{t\in[0,T]} \mathcal{W}_1(\underbrace{\Phi(\nu_t)}_{=\mu_t},\Phi(\tilde\nu_t)) \\
&= \sup_{t\in[0,T]}\sup_{\|\varphi\|_{\mathrm{Lip}}\le 1}\int_{\mathbb{R}^6}\varphi(z)\big(\,\mathrm{d}Z^\nu(t,0,\cdot)_\#\mu_0(z)-\mathrm{d}Z^{\tilde\nu}(t,0,\cdot)_\#\mu_0(z)\,\big) \\
&= \sup_{t\in[0,T]}\sup_{\|\varphi\|_{\mathrm{Lip}}\le 1}\int_{\mathbb{R}^6}\varphi(Z^\nu(t,0,z))-\varphi(Z^{\tilde\nu}(t,0,z))\,\mathrm{d}\mu_0(z) \\
&\le \sup_{t\in[0,T]}\int_{\mathbb{R}^6}Z^\nu(t,0,z)-Z^{\tilde\nu}(t,0,z)\,\mathrm{d}\mu_0(z)\le\alpha(\nu,\tilde\nu,z) \\
&\le \frac{\|\nabla K\|_{L^\infty(\mathbb{R}^3)}}{1-(1+\|\nabla K\|_{L^\infty(\mathbb{R}^3)})\cdot T}\tilde{\mathcal{W}}_1\Big((\nu_t)_{t\in[0,T]},(\tilde\nu_t)_{t\in[0,T]}\Big).
\end{aligned}$$

By our choice of T we showed the desired thesis. □

Theorem 5.11 (Dobrushin, 1979). (1) For any $T>0$ and $\mu_0\in\mathcal{P}_1(\mathbb{R}^6)$ there exists a unique solution $(\mu_t)_{t\in[0,T]}\in\mathfrak{m}_{T,\mu_0}$ to the Cauchy-Problem

$$\begin{cases}\dfrac{\mathrm{d}}{\mathrm{d}t}\mu_t(h)=\mu_t(G^\mu(t,\cdot)\cdot\nabla_z h)\\ \mu_t(h)\Big|_{t=0}=\mu_0(h)\end{cases}\qquad z=(x,v)\in\mathbb{R}^3\times\mathbb{R}^3,\quad G^\mu(t,z)=\begin{bmatrix}v\\E_{\mu_t}(x)\end{bmatrix}\tag{5.6}$$

in the sense of definition 5.2.

(2) If $\mu_0\in\mathcal{P}_1(\mathbb{R}^6)$ is absolutely continuous with respect to $\mathcal{L}^6$ with density $f_0\in C^1(\mathbb{R}^6)$ the solution $(\mu_t)_{t\in[0,T]}\in\mathfrak{m}_{T,\mu_0}$ to (5.6) has densities $(f_t)_{t\in[0,T]}$ where $f_t\in C^1(\mathbb{R}^6)$ and where

$$\begin{cases}\partial_t f_t+\tilde G^f(t,\cdot)\cdot\nabla_z f=0\\ f(t,z)\Big|_{t=0}=f_0(z)\end{cases}\qquad\text{where }\tilde G^f(t,z)=\begin{bmatrix}v\\ \int_{\mathbb{R}^3}K(x-y)\int_{\mathbb{R}^3}f(t,y,v)\,\mathrm{d}v\,\mathrm{d}y\end{bmatrix}$$

and $z=(x,v)\in\mathbb{R}^3\times\mathbb{R}^3$.

Proof. We only prove the first part, the second one is left as an exercise to the reader (Hint: Use Lemma 5.13). By the Lemmata 5.8 and 5.10 there exists some $\tilde T\le T$ where Φ contracts on the complete metric space $(\mathfrak{m}_{\tilde T,\mu_0},\tilde{\mathcal{W}}_1)$. Using Banachs fixed point theorem we infer the existence of a unique fixpoint of Φ,

$$(\nu_t)_{t\in[0,\tilde T]}=\Phi((\nu_t)_{t\in[0,\tilde T]})=(\mu_t)_{t\in[0,\tilde T]},\tag{5.7}$$

where $(\mu_t)_{t\in[0,\tilde T]}\in\mathfrak{m}_{\tilde T,\mu_0}$ is a solution of

$$\begin{cases}\dfrac{\mathrm{d}}{\mathrm{d}t}\mu_t(h)=\mu_t(G^\nu(t,\cdot)\cdot\nabla_z h)\\ \mu_t(h)\Big|_{t=0}=\mu_0(h)\end{cases}.$$

associated to $(\nu_t)_{t\in[0,\tilde{T}]}$ and μ_0 such that

$$[0,\tilde{T}] \to \mathbb{R} \quad t \mapsto \int_{\mathbb{R}^6} h(z)\,\mathrm{d}\mu_t(z)$$

is a differentiable map for any testfunction $h \in C_c^\infty(\mathbb{R}^6)$. We have the following bound for the difference of the electric fields:

$$\sup_{t\in[0,\tilde{T}]} \sup_{x\in\mathbb{R}^3} |E_{\nu_t}(x) - E_{\mu_t}(x)| \leq \|\nabla K\|_{L^\infty(\mathbb{R}^3)} \tilde{\mathcal{W}}_1((\nu_t)_{t\in[0,\tilde{T}]}, (\mu_t)_{t\in[0,\tilde{T}]}) \overset{(5.7)}{=} 0.$$

This implies that

$$|\mu_t(G^\nu(t,\cdot)\cdot\nabla_z h) - \mu_t(G^\nu(t,\cdot)\cdot\nabla_z h)| \leq \int_{\mathbb{R}^6} |(E_{\nu_t}(x) - E_{\mu_t}(x))\cdot\nabla_v h(x,v)|\,\mathrm{d}\mu_t(x,v) = 0$$

which allows us to replace G^ν by G^μ in (5.6) thus $(\mu_t)_{t\in[0,\tilde{T}]}$ is a solution to the Vlasov equation (5.6). The uniqueness of a solution is proven in complete analogy to the proof of lemma 5.6. Now we consider the unique solution $(\mu_t)_{t\in[0,\tilde{T}]}$ to the Vlasov equation up to $\tilde{T}$ and $(\tilde{\mu}_t)_{t\in[0,\tilde{T}]}$ with initial datum $\tilde{\mu}_t(h) = \mu|_{t=0} = \mu_{\tilde{T}/2}(h)$ for any $h \in C_c^\infty(\mathbb{R}^6)$. Since $(\tilde{\mu}_{t-\tilde{T}/2})_{t\in[\tilde{T}/2,3\tilde{T}/2]}$ satisfies the Vlasov equation with $\tilde{\mu}_{t-\tilde{T}/2}\Big|_{t=\tilde{T}/2} \equiv \mu_{\tilde{T}/2}$ we have

$$\mu_t \equiv \tilde{\mu}_{t-\frac{\tilde{T}}{2}} \quad \text{for any } t \in \left[\frac{\tilde{T}}{2}, \tilde{T}\right]$$

by the uniqueness of such a solution. By defining

$$\tilde{\tilde{\mu}}_t = \begin{cases} \mu_t, & t\in[0,\tilde{T}] \\ \tilde{\mu}_{t-\frac{\tilde{T}}{2}}, & t\in\left[\tilde{T}, \frac{2}{3}\tilde{T}\right] \end{cases}$$

we extend the solution from $[0,\tilde{T}]$ to $[0,3\tilde{T}/2]$. Since we can repeat this process until we covered the whole domain $[0,T]$ we thus obtain the existence and the uniqueness of a solution to the Vlasov equation. □

Remark 5.12. In this proof it is very important that the constant $\tilde{T}$ does not depend on the initial datum. We only assumed that the initial datum μ_0 is a probability measure on $(\mathbb{R}^6, \mathcal{B}(\mathbb{R}^6))$ to simplify the estimates. The result however can easily be generalized to positive Borel measures $\mu_0 \geq 0$ with

$$\int_{\mathbb{R}^6} (1+|z|)\,\mathrm{d}\mu_0(z) < +\infty.$$

which means that the measure has to be finite and have a finite first moment. △

Lemma 5.13. Let $(\mu_t)_{t\in[0,T]}$ be a solution to the Vlasov equation (5.6) and let Z^μ be the usual characteristics flow i.e.,

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t} Z^\mu(t,0,z) = G^\mu(t, Z^\mu(t,0,z)) \\ Z^\mu(0,0,\cdot) \equiv \mathrm{id} \end{cases}.$$

Then $\mu_t = Z^\mu(t,0,\cdot)_{\#}\mu_0$.

Proof. Let $\tilde{\mu}_t = Z^{\mu}(t, 0, \cdot)_{\#}\mu_0$. Then $\tilde{\mu}_t|_{t=0}| = \mu_0$ and $(\tilde{\mu}_t)_{t\in[0,T]}$ solves the respective Vlasov equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\mu}_t(h) = \mu_t(G^{\tilde{\mu}}(t,\cdot) \cdot \nabla_z h), \quad \text{for all } h \in C_c^\infty(\mathbb{R}^6).$$

Since the solutions to the Vlasov equation are unique the claim immediately follows. □

5.3. Stability of the Initial Datum

Lemma 5.14 (Grönwall)**.** Let $J := [t_0, t_1)$ where t_0, t_1 are two real numbers and $\alpha, \beta \in C^0(J, \mathbb{R})$.

(1) Let $\varphi \in C^1(J, \mathbb{R})$ such that $\dot{\varphi}(t) \leq \alpha(t) + \beta(t)\varphi(t)$ for any $t \in J$ then

$$\varphi(t) \leq \varphi(t_0) \exp\left\{\int_{t_0}^t \beta(\tau)\,\mathrm{d}\tau\right\} + \int_{t_0}^{t_1} \alpha(s) \cdot \exp\left\{\int_s^t \beta(\tau)\,\mathrm{d}\tau\right\} \mathrm{d}s\,.$$

(2) If $\varphi \in C^0(J, \mathbb{R})$, $\beta(t) \geq 0$ for any $t \in J$ and

$$\varphi(t) \leq \alpha(t) + \int_{t_0}^t \beta(s)\varphi(s)\,\mathrm{d}s \quad \text{for any } t \in J$$

then

$$\varphi(t) \leq \alpha(t) + \int_{t_0}^t \alpha(s)\beta(s) \cdot \exp\left\{\int_s^t \beta(\tau)\,\mathrm{d}\tau\right\} \mathrm{d}s\,.$$

Proof. (1) Define

$$\gamma(t) := \exp\left\{-\int_{t_0}^t \beta(\tau)\,\mathrm{d}\tau\right\}.$$

Then it holds that $\gamma(t) > 0$ for all t and $\gamma(t_0) = 1$. Moreover we deduce

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\varphi(t)\gamma(t)\Big) = \dot{\varphi}(t)\gamma(t) - \gamma(t)\varphi(t)\beta(t) \leq \alpha(t)\gamma(t)$$

thus by integrating and using the monotonicity of the integral we obtain

$$\gamma(t)\varphi(t) \leq \varphi(t_0) + \int_{t_0}^t \alpha(s)\gamma(s)\,\mathrm{d}s\,,$$

which gives the first inequality after multiplying with $\frac{1}{\gamma(t)}$.

(2) Setting

$$v(t) := \int_{t_0}^t \beta(s)\varphi(s)\,\mathrm{d}s$$

and using $\beta \geq 0$ and the assumed inequality for φ we obtain

$$\dot{v}(t) = \beta(t)\varphi(t) \leq \beta(t)\alpha(t) + \beta(t)v(t)$$

and $v(t_0) = 0$ so we can use the first part to see

$$\begin{aligned} v(t) &\leq \underbrace{v(t_0)}_{=0} \exp\left\{\int_{t_0}^t \beta(\tau)\,\mathrm{d}\tau\right\} + \int_{t_0}^t \alpha(s)\beta(s) \exp\left\{\int_s^t \beta(\tau)\,\mathrm{d}\tau\right\} \\ &= \int_{t_0}^t \alpha(s)\beta(s) \cdot \exp\left\{\int_s^t \beta(t)\,\mathrm{d}\tau\right\} \mathrm{d}s\,. \end{aligned}$$

Using $\varphi(t) \leq \alpha(t) + v(t)$ from the assumption for any t we obtain the desired inequality which concludes this proof. $\square$

Theorem 5.15. Let $K \in C_b^1(\mathbb{R}^3, \mathbb{R}^3)$, $T > 0$ and $\mu_0, \nu_0 \in \mathcal{P}_1(\mathbb{R}^6)$ and $(\mu_t)_{t\in[0,T]}, (\nu_t)_{t\in[0,T]}$ be the solutions to the Vlasov equation (5.6) with initial datum μ_0 and μ_0 respectively given by theorem 5.11. Then there exists a constant $C > 0$ which is independent of T such that

$$\mathcal{W}_1(\mu_t, \nu_t) \leq C\mathrm{e}^{Ct}\mathcal{W}_1(\mu_0, \nu_0) \quad \text{for any } t \in [0,T].$$

Proof. By lemma 5.13 the solutions to (5.6) are given by $\mu_t = Z^\mu(t,0,\cdot)_\#\mu_0$ thus

$$\begin{aligned}
\mathcal{W}_1(\mu_t, \nu_t) &= \mathcal{W}_1\Big(Z^\mu(t,0,\cdot)_\#\mu_0, Z^\nu(t,0,\cdot)_\#\nu_0\Big)\\
&\leq \mathcal{W}_1\Big(Z^\mu(t,0,\cdot)_\#\mu_0, Z^\mu(t,0,\cdot)_\#\nu_0\Big) + \mathcal{W}_1\Big(Z^\mu(t,0,\cdot)_\#\nu_0, Z^\nu(t,0,\cdot)_\#\nu_0\Big)\\
&= \sup_{\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^6)}\leq 1} \int_{\mathbb{R}^6} \varphi(Z^\mu(t,0,z)(\,\mathrm{d}\mu_0 - \mathrm{d}\nu_0\,)(z)\\
&\quad + \sup_{\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^6)}\leq 1} \int_{\mathbb{R}^6} \varphi(Z^\mu(t,0,z)) - \varphi(Z^\nu(t,0,z,))\,\mathrm{d}\nu_0(z)\\
&=: \mathrm{I} + \mathrm{II}
\end{aligned}$$

We first consider the term I: Using that K has a bounded derivative we obtain that K is Lipschitz continuous with

$$|K(x) - K(y)| \leq \|\nabla K\|_{L^\infty(\mathbb{R}^3)}|x-y|,$$

analogous to the proof of lemma 5.8. This gives the following estimate:

$$|E_{\mu_t}(x) - E_{\mu_t}(y)| \leq \int_{\mathbb{R}^6} |K(x-u) - K(y-v)|\,\mathrm{d}\mu_t(u,v) \leq \|\nabla K\|_{L^\infty(\mathbb{R}^3)}|x-y|$$

and

$$\begin{aligned}
&|G^\mu(t, Z^\mu(t,0,z)) - G^\mu(t, Z^\mu(t,0,z'))|\\
&= \Big(|V^\mu(t,0,z) - V^\mu(t,0,z')|^2 + |E_{\mu_t}(X^\mu(t,0,z) - E_{\mu_t}(X^\mu(t,0,z'))|^2\Big)^{1/2}
\end{aligned} \tag{5.8}$$

where $Z^\mu = \begin{bmatrix} V^\mu & X^\mu\end{bmatrix}^\top$ is the flow of G^ν. The functions $z \mapsto V^\nu(t,0,z)$ and $z \mapsto E_{\mu_t}(X^\mu(t,0,z))$ are both Lipschitz continuous with Lipschitz constants 1 and $\|K\|_{L^\infty(\mathbb{R}^3)}$ respectively. From (5.8) we thus get the bound

$$\big|G^\mu(t, Z^\mu(t,0,z)) - G^\mu(t, Z^\mu(t,0,z'))\big| \leq \max\Big\{1, \|K\|_{L^\infty(\mathbb{R}^3)}\Big\}\big|Z^\mu(t,0,z) - Z^\mu(t,0,z')\big|.$$

Consider

$$\begin{aligned}
&|Z^\mu(t,0,z) - Z^\mu(t,0,z')|\\
&\qquad \leq \underbrace{|Z^\mu(0,0,z) - Z^\mu(0,0,z')|}_{=|z-z'|} + \int_0^t \frac{\mathrm{d}}{\mathrm{d}s}\Big(Z^\mu(s,0,z) - Z^\mu(s,0,z')\Big)\,\mathrm{d}s\\
&\qquad = |z-z'| + \int_0^t G^\mu(s, Z^\mu(s,0,z)) - G^\mu(s, Z^\mu(s,0,z'))\,\mathrm{d}s\\
&\qquad \leq |z-z'| + \underbrace{\max\Big\{1, \|\nabla K\|_{L^\infty(\mathbb{R}^3)}\Big\}}_{=:\tilde{C}} \int_0^t Z^\mu(s,0,z) - Z^\mu(s,0,z')\,\mathrm{d}s\,,
\end{aligned}$$

so by Grönwalls lemma (Lemma 5.14) it holds that

$$\begin{aligned} |Z^\mu(t,0,z) - Z^\mu(t,0,z')| &\leq |z-z'|\left(1 + \int_0^1 \tilde{C} \exp\left\{\int_s^t \tilde{C}\,\mathrm{d}\tau\right\} \mathrm{d}s\right) \\ &\leq |z-z'| \cdot C\mathrm{e}^{Ct}. \end{aligned}$$

We get the same bound if we compose Z^μ with some Lipschitz continuous function φ satisfying $\|\varphi\|_{\mathrm{Lip}(\mathbb{R}^6)} \leq 1$ therefore $\mathrm{I} \leq \mathrm{e}^{Ct}\mathcal{W}_1(\mu_0,\nu_0)$. Using similar estimates as in the proof of lemma 5.10 gives

$$\begin{aligned} |Z^\mu(t,0,z) - Z^\nu(t,0,z)| &= \int_0^t \left|G^\mu(s,Z^\mu(s,0,z)) - G^\nu(s,Z(s,0,z))\right| \mathrm{d}s \\ &\leq 2\big(1 + \|\nabla K\|_{L^\infty(\mathbb{R}^3)}\big) \int_0^t |Z^\mu(s,0,z+Z^\nu(s,0,z)| + \mathcal{W}_1(\mu_s,\nu_s)\,\mathrm{d}s \end{aligned}$$

So for $\lambda(t) := \int_{\mathbb{R}^6} |Z^\mu(t,0,z) - Z^\nu(t,0,z)|\,\mathrm{d}\nu_0(z)$ this leads to

$$\lambda(t) \leq C\int_0^t \lambda(s) + \mathcal{W}_1(\mu_s,\nu_s)\,\mathrm{d}s \tag{5.9}$$

and since we are dealing with Lipschitzfunctions with Lipschitz constant smaller than one we get the following estimate for II:

$$\mathrm{II} \leq \lambda(t) \overset{(5.9)}{\leq} \int_0^t \lambda(s) + \mathcal{W}_1(\mu_s,\nu_s)\,\mathrm{d}s\,.$$

Combining the estimates for I and II one finds

$$\mathcal{W}_1(\mu_t,\nu_t) \leq \mathrm{e}^{Ct}\mathcal{W}_1(\mu_0,\nu_0) + C\int_0^t \lambda(s) + \mathcal{W}_1(\mu_s,\nu_s)\,\mathrm{d}s$$

and using (5.9)

$$\mathcal{W}_1(\mu_t,\nu_t) + \lambda(t) \leq \mathrm{e}^{Ct}\mathcal{W}_1(\mu_0,\nu_0) + C\int_0^t \lambda(s)\mathcal{W}_1(\mu_s,\nu_s)\,\mathrm{d}s$$

thus by lemma 5.14 we find using that $\lambda(t) \geq 0$

$$\begin{aligned} \mathcal{W}_1(\mu_t,\nu_t) \leq \mathcal{W}_1(\mu_t,\nu_t) + \lambda(t) &\leq \mathrm{e}^{Ct}\mathcal{W}_1(\mu_0,\nu_0) + \int_0^t \mathrm{e}^{Cs}\mathcal{W}_1(\mu_0,\nu_0)C\mathrm{e}^{C(t-s)}\,\mathrm{d}s \\ &\leq C\mathrm{e}^{Ct}\mathcal{W}_1(\mu_0,\nu_0). \end{aligned}$$

This shows the claim. □

6. The Vlasov Equation: Derivation

We consider $\{(x_i, v_i)\}_{i=1}^N \subset \mathbb{R}^3 \times \mathbb{R}^3$ moving particles in the phase space, each of them having mass one. Using $F = m \cdot a$,

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t} x_i(t) = v_i(t) \\ \dfrac{\mathrm{d}}{\mathrm{d}t} v_i(t) = \displaystyle\sum_{\substack{j=1 \\ j \neq i}}^N K(x_i(t) - x_j(t)) \end{cases}. \tag{6.1}$$

Newton's third law ensures that $K(x) = -K(-x)$ for any $x \in \mathbb{R}^3$ thus $K(0) = 0$ which makes the condition $j \neq i$ in the sum unnecessary. We are interested in N very large (something like the amount of molecules in air etc). The Goal is to analyze the behaviour of (6.1) for very large N.

Note that $\frac{\mathrm{d}}{\mathrm{d}t} v_i(t) = \mathcal{O}(N)$ if K is bounded. We thus have to consider the right scaling and the right time to observe the change of the system. In order to do that we want the size of the system to be one. We re-scale in time by setting $\tilde{t} := \sqrt{N} t$. If $\tilde{t}$ is of order one we obtain that $t = \mathcal{O}(\frac{1}{\sqrt{N}})$ i.e., we are going to slower times. Moreover we get

$$\tilde{x}_i(t) = x_i\left(\frac{t}{\sqrt{N}}\right) \quad \text{and} \quad \tilde{v}_i(t) = \frac{1}{\sqrt{N}} v_i\left(\frac{t}{\sqrt{N}}\right).$$

This leads to a slower parameterization of the curves and thus to smaller velocities:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} x_i(t) &= \sqrt{N} \frac{\mathrm{d}}{\mathrm{d}\tilde{t}} \tilde{x}_i(\tilde{t}) \\ \frac{\mathrm{d}}{\mathrm{d}t} v_i(t) &= N \frac{\mathrm{d}}{\mathrm{d}\tilde{t}} \tilde{v}_i(\tilde{t}) \\ x_i(t) &= \tilde{x}_i(\tilde{t}) \quad \text{and} \\ v_i(t) &= \sqrt{N} \cdot \tilde{v}_i(\tilde{t}). \end{aligned}$$

Plugging this into (6.1) and dividing by N and removing all the tildes we obtain the new system

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t} x_i(t) = v_i(t) \\ \dfrac{\mathrm{d}}{\mathrm{d}t} v_i(t) = \dfrac{1}{N} \displaystyle\sum_{j=1}^N K(x_i(t) - x_j(t)) \end{cases}. \tag{6.2}$$

We will use this system as a starting of point for our derivations. One possibility to undesertand how one can derive the Vlasov-equation is to look at the empirical measure of the trajectories:

Definition 6.1. Let $N \in \mathbb{N}$ and consider $Z_N := (z_i)_{i=1}^N$ where $z_i = (x_i, v_i)$ are our particles as before. We define the **empirical measure** by

$$\mu_{Z_N} := \frac{1}{N} \sum_{j=1}^{N} \delta_{Z_j}$$

which is a probability measure on $(\mathbb{R}^6, \mathcal{B}(\mathbb{R}^6))$.

This measure gives the relative frequency of finding a particle at a given phase space position. At time $t = 0$ we consider a sequence of initial conditions $(Z_N)_{N\in\mathbb{N}}$ such that

$$\mu_{Z_N} \rightharpoonup f_0 \, \mathrm{d}\mathcal{L}^6$$

and such that $Z_N(t) := (x_i(t), v_i(t))_{i=1}^N$ are the respective solutions to (6.2). We want the following diagram to commute:

$$\begin{tikzcd}
\mu_{Z_N} \arrow[r, "N\to+\infty"] \arrow[d, "(6.2)"] & f_0 \, \mathrm{d}\mathcal{L}^6 \arrow[d, "(5.1)"] \\
\mu_{Z_N(t)} \arrow[r, "N\to+\infty"] & f_t \, \mathrm{d}\mathcal{L}^6
\end{tikzcd}.$$

Theorem 6.2. Let $K \in C_b^0(\mathbb{R}^3)$ such that $K(x) = -K(-x)$. Let $T > 0$ and $N \in \mathbb{N}$. Then:

(1) For any initial datum $Z_N^{\mathrm{in}} = (x_i^{\mathrm{in}}, v_i^{\mathrm{in}})_{i=1}^N \in \mathbb{R}^{6N}$ the N-particle ODE (6.2) with initial datum Z_N^{in} has a solution $Z_N(t) = (z_i(t))_{i=1}^N$ in $C^1([0,T]; \mathbb{R}^6)$.

(2) The empirical measure
$$\mu_{Z_N(t)} = \frac{1}{N} \sum_{j=1} \delta_{Z_j(t)}$$
is the unique solution of the Vlasov equation from Theorem 5.11 with initial datum
$$\mu_0 = \mu_{Z_N^{\mathrm{in}}} = \frac{1}{N} \sum_{j=1}^{N} \delta_{Z_j^{\mathrm{in}}}.$$

Proof. (1) This follows directly from the Cauchy-Lipschitz theory using that K is Lipschitz continuous on $\mathbb{R}^3$.

(2) Take a testfunction $h \in C_c^\infty(\mathbb{R}^6)$ and see that

$$\int_{\mathbb{R}^6} h \, \mathrm{d}\mu_{Z_N(t)} = \frac{1}{N} \sum_{j=1}^{N} (h(z_j(t)).$$

As $t \mapsto Z_N(t)$ is a differentiable function, the mapping

$$t \in [0,T] \mapsto \int_{\mathbb{R}^6} h \, \mathrm{d}\mu_{Z_N(t)}$$

is continuosly differentiable on $[0,T]$. Moreover

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\mu_{Z_N(t)}(h) &= \frac{1}{N}\sum_{j=1}^{N}\frac{\mathrm{d}}{\mathrm{d}t}h(z_j(t))\\
&= \frac{1}{N}\sum_{j=1}^{N}(\partial_{x_j}h)(z_j(t))\cdot v_j(t) + \frac{1}{N}\sum_{j=1}^{N}(D_v h)(z_j(t))\cdot\frac{1}{N}\sum_{n=1}^{N}K(x_j(t)-x_n(t))\\
&= \frac{1}{N}\sum_{j=1}^{N}(\partial_{x_j}h)(z_j(t))\cdot v_j(t) + \frac{1}{N}\sum_{j=1}^{N}(D_v h)(z_j(t))\cdot\int_{\mathbb{R}^6}K(x_j(t)-y)\,\mathrm{d}\mu_{Z_n(t)}(y,v)\\
&= E_{\mu_{Z_N(t)}}(x_j(t)).
\end{aligned}
$$

One thus sees that

$$
\begin{cases}
\dfrac{\mathrm{d}}{\mathrm{d}t}\mu_{Z_N(t)}(h) = \mu_{Z_N(t)}(v\cdot\nabla_x h + E_{\mu_{Z_N(t)}})\cdot\nabla_x h)\\
\mu_{Z_N(t)}\Big|_{t=0} = \mu_{Z_N^{\mathrm{in}}(t)}
\end{cases}.
$$

Moreover note $\mu_{Z_N(t)}(\mathbb{R}^6) = 1$ and

$$
\int_{\mathbb{R}^6}|z|\,\mathrm{d}\mu_{Z_N(t)}(z) = \frac{1}{N}\sum_{j=1}^{N}|z_j(t)| < +\infty
$$

which implies that $\mu_{Z_N(t)} \in \mathcal{P}_1(\mathbb{R}^6)$ for any $t \in [0,T]$. Using that $Z_N(t)$ and K are differentiable functions and $K \in C_b^1(\mathbb{R}^3)$ we deduce that the mapping

$$
[0,T] \to \mathbb{R}^3, \quad t \mapsto E_{\mu_{Z_N(t)}}(x) = \frac{1}{N}\sum_{j=1}^{N}K(x - x_j(t))
$$

is continuous for any $x \in \mathbb{R}^3$ and therefore $\mu_{Z_N(t)} \in \mathfrak{m}_{T,\mu_0}$. By theorem 5.11 the claim follows. □

Theorem 6.3. Assume $K \in C_b^1(\mathbb{R}^3)$ with K odd, that is $K(x) = -K(-x)$ for any $x \in \mathbb{R}^3$. Moreover let $T > 0$ and $f_0 \in C^0(\mathbb{R}^6)$ be the a probability density function on $\mathbb{R}^6$ meaning that f_0 is nonnegative and $\|f_0\|_{L^1} = 1$. We moreover require the expectation to exists i.e.,

$$
\int_{\mathbb{R}}|z\cdot f_0(z)|\,\mathrm{d}z < +\infty.
$$

If the above conditions are met the following holds:

(1) The Cauchy problem (5.4) with initial datum

$$
\mu_0(B) = \int_B f_0(z)\,\mathrm{d}z
$$

has a unique solution given by

$$
\mu_t(B) = \int_B f_t(z)\,\mathrm{d}z\,, \quad \text{where} \quad f_t = f_0(Z^\mu(0,t,\cdot)).
$$

(2) For any $N \in \mathbb{N}$ we take a vector with initial conditions $Z_N^{\text{in}} = (Z_{i,N}^{\text{in}})_{i=1}^N \in \mathbb{R}^{6N}$ such that

$$\mu_{Z_N^{\text{in}}} \frac{1}{N} \sum_{j=1}^{N} \delta_{Z_{j,N}^{\text{in}}} \quad \text{satisfies} \quad \mathcal{W}_1\left(\mu_{Z_N^{\text{in}}}, f_0 \, \mathrm{d}\mathcal{L}^6\right) \xrightarrow{N \to +\infty} 0,$$

where $f_0 \, \mathrm{d}\mathcal{L}^6$ denotes the probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}^6)$ with density f_0 (or rather $\frac{\mathrm{d}f_0}{\mathrm{d}\mathcal{L}^6}$). Moreover let $Z_N(t) = (Z_{i,N}(t))_{i=1}^N$ the solution of (6.2) with initial datum Z_N^{in}. Then for any $t \in [0,T]$ it holds that

$$\mathcal{W}_1\left(\mu_{Z_N(t)}, f_t \, \mathrm{d}\mathcal{L}^6\right) \xrightarrow{N \to +\infty} 0$$

which gives us by Corollary 4.10 that $\mu_{Z_N(t)} \rightharpoonup f_t \, \mathrm{d}\mathcal{L}^6$.

Proof. (1) The uniqueness and existence of the solutions can be deduced from theorem 5.11 and the fact that $\mu_t = f_t \, \mathrm{d}\mathcal{L}^6$ follows from $\mu_t = Z^\mu(t, 0, \cdot)_{\#}\mu_0$ by lemma 5.13.

(2) By theorem 6.2 we obtain that $\mu_{Z_N(t)}$ is a solution of the Vlasov equation (5.1). Using theorem 5.15 we obtain

$$\mathcal{W}_1\left(\mu_{Z_N(t)}, f_t \, \mathrm{d}\mathcal{L}^6\right) \le C\mathrm{e}^{Ct} \mathcal{W}_1\left(\mu_{Z_N^{\text{in}}}, f_0 \, \mathrm{d}\mathcal{L}^6\right) \xrightarrow{n \to +\infty} 0.$$

This gives the desired convergence and concludes the proof. □

Remark 6.4.
- If $f_0 \in C^0(\mathbb{R}^6)$ one can show that f_t is then a classical solution.
- By theorem 6.3 it remains to show that one can generate N-tuples $Z_N^{\text{in}} \in \mathbb{R}^{6N}$ such that

$$\mu_{Z_N^{\text{in}}} \frac{1}{N} \sum_{j=1}^{N} \delta_{Z_{j,N}^{\text{in}}} \quad \text{satisfies} \quad \mathcal{W}_1\left(\mu_{Z_N^{\text{in}}}, f_0 \, \mathrm{d}\mathcal{L}^6\right) \xrightarrow{N \to +\infty} 0.$$

For details we refer to the lecture notes of Golse.

△

A. More Results in PDE-Theory

Theorem A.1. Let $S := \Big\{(t,x) \mid t \in [a,b], x \in \mathbb{R}^d\Big\}$ and assume that the function $b : S \to \mathbb{R}^d$ is Lipschitz continuous in x i.e.,

$$|b(t,x) - b(t,y)| \leq L|x-y|$$

for some constant $L > 0$ and for any $(t,x),(t,y) \in S$. Then for any $t_0 \in [a,b]$ the initial value problem

$$\begin{cases} \dot\gamma(t) = b(t,\gamma(t)) \\ \gamma(t_0) = \gamma_0 \end{cases}$$

has a unique solution.

Proof. For $u \in C^0([a,b],\mathbb{R}^d)$ we define

$$\Phi : C^0([a,b],\mathbb{R}^d) \to C^0([a,b],\mathbb{R}^d), \quad \Phi[u](t) := \gamma_0 + \int_{t_0}^t b(s,u(s))\,\mathrm{d}s\,, \quad t \in [a,b]$$

If γ is a fixed point of Φ then

$$\frac{\mathrm{d}}{\mathrm{d}t}\gamma(t) = \frac{\mathrm{d}}{\mathrm{d}t}\Phi[\gamma](t) = b(t,u(t))$$

and if γ solves the initial value problem because

$$\gamma(t) = \gamma(t_0) + \int_{t_0}^t \frac{\mathrm{d}}{\mathrm{d}s}\gamma(s)\,\mathrm{d}s = \gamma_0 + \int_{t_0}^t b(s,\gamma(s))\,\mathrm{d}s\,.$$

To use Banach's fixed point theorem one has to introduce a suitable norm:

$$|||\gamma||| := \sup_{t\in[a,b]} \Big\{\mathrm{e}^{-2Lx}|\gamma(t)|\Big\}.$$

We then find

$$\begin{aligned}
\mathrm{e}^{-2Lx}|\Phi[u](x) - \Phi[v](x)| &= \mathrm{e}^{-2Lx}\left|\int_{t_0}^t f(s,u(s)) - f(s,v(s))\,\mathrm{d}s\right| \\
&\leq \mathrm{e}^{-2Lx}\int_{t_0}^t |f(s,u(s)) - f(s,v(s))|\,\mathrm{d}s \\
&\leq L\mathrm{e}^{-2Lx}\int_{t_0}^t |u(s) - v(s)|\,\mathrm{d}s \\
&\leq L\mathrm{e}^{-2Lx}\int_{t_0}^t \mathrm{e}^{2Ls}\underbrace{\mathrm{e}^{-2Lt}|u(s)-v(s)|}_{\leq |||u-v|||}\,\mathrm{d}s \\
&\leq L\mathrm{e}^{-2Lx}|||u-v||| \underbrace{\int_{t_0}^t \mathrm{e}^{2Ls}\,\mathrm{d}s}_{=\left(\mathrm{e}^{2Lt}-\mathrm{e}^{2Lt_0}\right)/(2L)} \leq \frac{1}{2}|||u-v|||.
\end{aligned}$$

We thus find that Φ is a contraction:

$$|||\Phi[u] - \Phi[v]||| \leq \frac{1}{2}|||u - v|||.$$

From Banach's fixed point theorem we deduce the existence of a unique fixed point. Since all solutions are fixed points of Φ and vice versa we obtain the desired result. $\square$